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32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

<u>INSTRUMENTATION SUBMITTAL</u>

SECTION 17500 - INSTRUMENTATION

January 27, 2026

ENGINEER: WRA

CONTRACTOR: JOHNSTON CONSTRUCTION

MANUFACTURERS' REPRESENTATIVE FOR ENGINEERED INSTRUMENTATION, PROCESS EQUIPMENT AND PUMPING SYSTEMS

Branch Offices: RICHMOND, VA and LEOLA, PA

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General Comments:

1. This submittal assumes cable lengths for the various instruments included. Contractor to confirm cable lengths for instruments.

Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION
1	Endress + Hauser	FMR20-CBPBMVCEVEE3+Z1	Radar Level Transmitter
2	Anchor Scientific	P60NO-SSTNM	Float Switch, NO
3	Endress + Hauser	RSG35-B2A+C1	Digital Chart Recorder
3	Endress + Hauser	71187780	1GB Neutral SD Memorycard
4	Endress + Hauser	RIA452-C211A11A	Digital Process Meter Display

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Spare Parts Index:

SHT	MANUFACTURER	MODEL	DESCRIPTION	QTY
3	Endress + Hauser	71187780	1GB Neutral SD Memorycard	1

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Data Sheet 1

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 01/27/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Endress + Hauser: Radar Level Transmitter Model Number: FMR20-CBPBMVCEVEE3+Z1 Tags/Service: LT-01 / Pump Station Wet Well Specifications: • FMR20 - Micropilot FMR20 series time-of-flight radar measurement • CB - CSA C/US IS CI.I Div.2 Gr.A-d, AEx/Ex ia IIC T4 approvals • P - 2 wire power supply, 4-20mA HART output, HART/ Bluetooth (App) configuration operation • BM - 40mm/1-$\frac{1}{2}$" antenna, 15m liquid, −40°C…80°C max measuring range • VCE - Thread ASME MNPT1 process connection rear side, PVDF material, FNPT $\frac{1}{2}$ conduit connection • VEE - Thread ASME MNPT1-$\frac{1}{2}$ process connection front side, PVDF material • 3 - 20m/65ft cable length • Z1 - Tagging

TI01267F/00/EN/08.20
71492916
2020-08-17

RADAR LEVEL TRANSMITTER

TAGS: LT-01

Technical Information
Micropilot FMR20 HART

Free space radar

Level measurement for liquids

Application

- Ingress protection: IP66/68 / NEMA 4X/6P
- Measuring range: up to 20 m (66 ft)
- Process temperature: −40 to 80 °C (−40 to 176 °F)
- Process pressure: −1 to 3 bar (−14 to 43 psi)
- Accuracy: up to ± 2 mm (0.08 in)
- International explosion protection certificates

Your benefits

- Level measurement for liquids in storage tanks, open basins, pump shafts and canal systems
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Simple, safe and secure wireless remote access – ideal for installation in hazardous areas or places difficult to reach
- Commissioning, operation and maintenance via free iOS / Android app SmartBlue – saves time and reduces costs
- Full PVDF body - for a long sensor lifetime
- Hermetically sealed wiring and fully potted electronics – eliminates water ingress and allows operation under harsh environmental conditions
- Most compact radar due to unique radar chip design – fits in limited space installations
- Best price-performance-ratio radar

Endress+Hauser
People for Process Automation

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Important document information

Symbols used

Symbols for certain types of information and graphics

Permitted
Procedures, processes or actions that are permitted

Forbidden
Procedures, processes or actions that are forbidden

Tip
Indicates additional information

Reference to documentation

Reference to graphic

▶
Notice or individual step to be observed

1., 2., 3.
Series of steps

↳
Result of a step

1, 2, 3, ...
Item numbers

A, B, C, ...
Views

Terms and abbreviations

BA
Document type "Operating Instructions"

KA
Document type "Brief Operating Instructions"

TI
Document type "Technical Information"

SD
Document type "Special Documentation"

XA
Document type "Safety Instructions"

PN
Nominal pressure

MWP
MWP (Maximum working pressure/max. process pressure)
The MWP can also be found on the nameplate.

ToF
Time of Flight

FieldCare
Scalable software tool for device configuration and integrated plant asset management solutions

DeviceCare
Universal configuration software for Endress+Hauser HART, PROFIBUS, FOUNDATION Fieldbus and Ethernet field devices

DTM
Device Type Manager

ε_r (Dk value)
Relative dielectric constant

Operating tool

The term "operating tool" is used in place of the following operating software:
- FieldCare / DeviceCare, for operation via HART communication and PC
- SmartBlue (app), for operation using an Android or iOS smartphone or tablet

BD
Blocking Distance; no signals are analyzed within the BD.

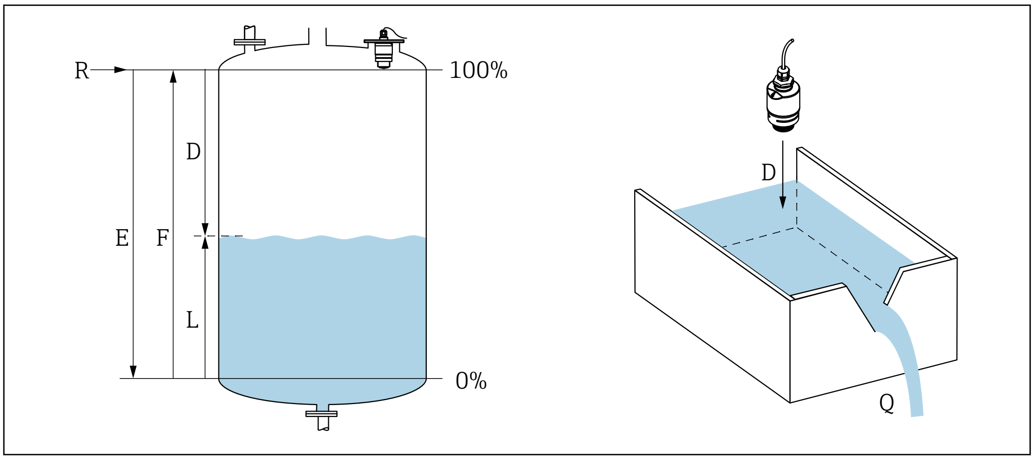
PLC
Programmable logic controller (PLC)

Product life cycle

Engineering	▪ Proven radar measuring technology ▪ Level measurement and open channel flow measurement for Ex and non-Ex areas ▪ Flooding detection ▪ Wide range of installation possibilities and accessories ▪ Highest degree of ingress protection ▪ 2D/3D drawings ▪ Spec Sheet Producer ▪ Applicator Selection tool for selecting the perfect measurement solution Device not compatible with transmitters and sensors that use ultrasonic measurement technology (e.g. Prosonic FMU9x, FDU9x)
Procurement	▪ Best price-performance-ratio radar ▪ Global availability ▪ Order code includes variety of mounting accessories and decentralized RIA15 process indicator for HART
Installation	▪ Rear and front thread for flexible installation ▪ Slip-on flange for nozzle installation ▪ Complete measuring point: including mounting accessory, RIA15 and flooding protection tube
Commissioning	▪ Quick and easy setup with the SmartBlue app and DeviceCare / FieldCare or RIA15 ▪ No additional tools or adapters required ▪ Local languages (up to 15)
Operation	▪ Continuous self-monitoring ▪ Diagnostics information according to NAMUR NE107 with remedial measures in the form of plain text messages ▪ Signal curve via SmartBlue (app) and DeviceCare / FieldCare ▪ Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via *Bluetooth*® wireless technology
Maintenance	▪ No maintenance required ▪ Technical experts on-call around the world
Retirement	▪ Environmentally responsible recycling concepts ▪ RoHS compliance (restriction of certain hazardous substances), lead-free soldering of electronic components

Measuring principle

The Micropilot is a "downward-looking" measuring system, which functions according to the time-of-flight (ToF) method. It measures the distance from the reference point **R** to the product surface. Radar pulses are emitted by an antenna, reflected off the product surface and received again by the radar system.



1 *Level / flow calibration parameter*

E Empty calibration (= zero)
F Full calibration (= span)
D Measured distance
L Level (L = E - D)
Q Flow rate at measuring weirs or channels (calculated from the level using linearization)
R Reference point

Input

The reflected radar pulses are received by the antenna and transmitted to the electronics. A microprocessor evaluates the signals and identifies the level echo caused by the reflection of the radar pulses at the product surface. This clear signal detection system benefits from over 30 years' experience with time-of-flight procedures.

The distance **D** to the product surface is proportional to the time of flight **t** of the pulse:

$D = c \cdot t/2$,

where **c** is the speed of light.

Based on the known empty distance **E**, the level **L** is calculated:

$L = E - D$

Output

The Micropilot is calibrated by entering the empty distance **E** (= zero point) and the full distance **F** (= span).

- Current output: 4 to 20 mA
- Digital output (HART, SmartBlue): 0 to 15 m (0 to 49 ft) [1] or 0 to 20 m (0 to 66 ft) depending on antenna version

1) order code 010 (approval) "GA", "GB", "GR": 0 to 10 m (0 to 32.8 ft)

Input

Measured variable

The measured variable is the distance between the reference point and the product surface.

The level is calculated based on **E**, the empty distance entered.

Measuring range

Maximum measuring range

- Device with 40 mm (1.5 in) antenna: 15 m (49 ft) [2)]
- ~~Device with 80 mm (3 in) antenna: 20 m (66 ft)~~

Installation requirements

- Recommended tank height > 1.5 m (5 ft) for media with low ε_r value
- Open channel minimum width 0.5 m (1.6 ft)
- Calm surfaces
- No agitators
- No buildup
- Relative dielectric constant $\varepsilon_r > 4$
  Contact Endress+Hauser for lower ε_r values

Usable measuring range

The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections.

Media groups
- $\varepsilon_r = 4$ to 10
  e.g. concentrated acid, organic solvents, ester, aniline, alcohol, acetone.
- $\varepsilon_r > 10$
  e.g. conductive liquids, aqueous solutions, diluted acids and bases

Reduction of the maximum possible measuring range by:
- Media with bad reflective properties (= low ε_r value)
- Formation of buildup, particularly of moist products
- Strong condensation
- Foam generation
- Freezing of sensor

Operating frequency

K-band (~ 26 GHz)

Transmission power

Mean power density in the direction of the beam
- At a distance of 1 m (3.3 ft): < 12 nW/cm^2
- At a distance of 5 m (16 ft): < 0.4 nW/cm^2

Output

Output signal

4 to 20 mA

An 4 to 20 mA interface is used for measured value output and to power to the device.

Digital output

HART®

- Signal encoding; FSK ±0.5 mA over current signal
- Data transmission rate;1 200 Bit/s

Bluetooth® wireless technology (available as an optional extra)

The device has a *Bluetooth*® wireless technology interface and can be operated and configured via this interface using the SmartBlue app.

2) order code 010 (approval) "GA", "GB", "GR": 10 m (32.8 ft)

- The range under reference conditions is 25 m (82 ft)
- Incorrect operation by unauthorized persons is prevented by means of encrypted communication and password encryption
- The *Bluetooth*® wireless technology interface can be deactivated

Signal on alarm

Depending on the interface, failure information is displayed as follows:
- Current output
  Alarm current: 22.5 mA (in accordance with NAMUR recommendation NE 43)
- Operating tool via digital communication (HART) or SmartBlue (app)
  - Status signal (as per NAMUR Recommendation NE 107)
  - Plain text display with remedial action

Linearization

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume. In DeviceCare and FieldCare, there are preprogrammed linearization tables for volume calculation in vessels.

Pre-programmed linearization curves
- Horizontal cylindrical tank
- Sphere
- Tank with pyramid bottom
- Tank with conical bottom
- Tank with flat bottom

Other linearization tables of up to 32 value pairs can be entered manually.

Protocol-specific data, HART

Manufacturer ID
17 (0x11)

Device type ID
44 (0x112c)

HART specification
7.0

Device description files (DTM)
Information and files under:
- www.endress.com
- www.hartcomm.org

HART load
Min. 250 Ω

HART device variables
Assignment of HART device variables is fixed and cannot be changed.
- **Measured values for PV (primary variable)**
  Level linearized
- **Advanced diag. measured values for SV (secondary variable)**
  Distance
- **Advanced diag. measured values for TV (tertiary variable)**
  Relative echo amplitude
- **Advanced diag. measured values for QV (quarternary variable)**
  Temperature

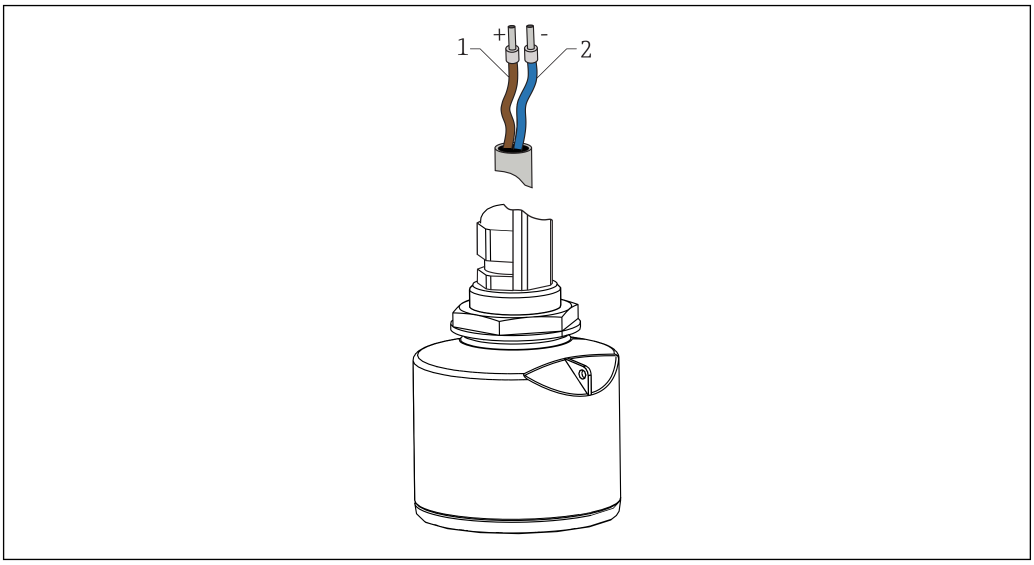
Supported functions
Additional transmitter status

Multidrop current
4 mA

Time for connection setup
< 1 s

Electrical connection

Cable assignment



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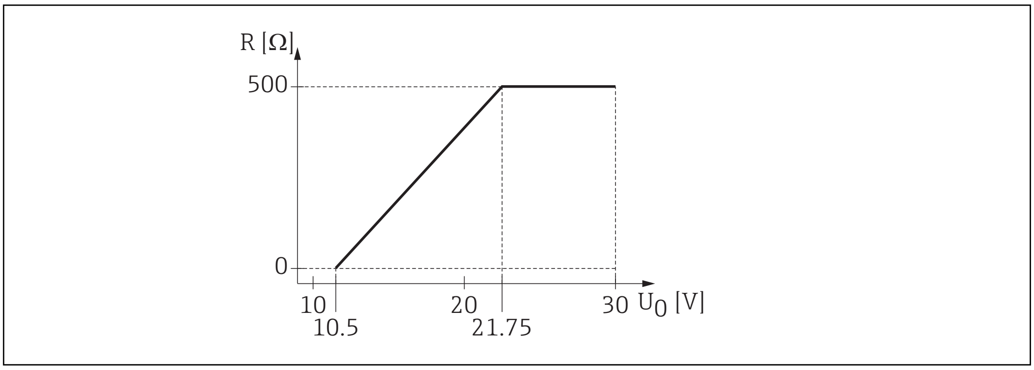
2 *Cable assignment*

1 *Plus, brown wire*
2 *Minus, blue wire*

Supply voltage

10.5 to 30 V_{DC}

An external power supply is necessary.



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3 *Maximum load R, depending on supply voltage U_0 of power supply unit*

Battery operation
The sensor's *Bluetooth*® wireless technology communication can be disabled to increase the operating life of the battery.

Potential equalization
No special measures for potential equalization are required.

Various power supply units can be ordered as an accessory from Endress+Hauser.

Power consumption

Maximum input power: 675 mW

Current consumption

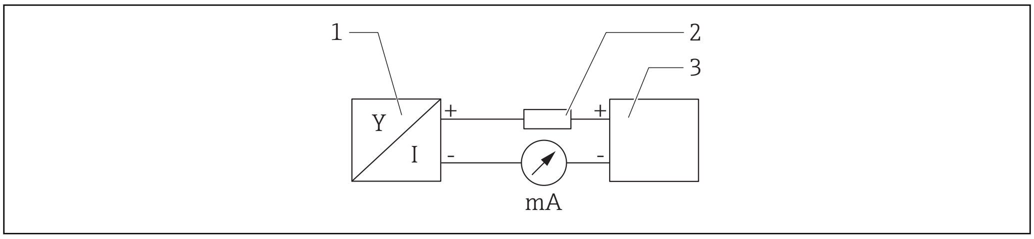
- Maximum input current: <25 mA
- Maximum start-up current: 3.6 mA

Starting time

First stable measured value after 20 s (with supply voltage = 24 V_{DC})

Power supply failure	The configuration remains stored in the sensor.
Connecting the device	**4 to 20 mA HART block diagram**

Connection of the device with HART communication, power source and 4 to 20 mA display



4 *Block diagram of HART connection*

1 *Device with HART communication*
2 *HART resistor*
3 *Power supply*

The HART communication resistor of 250 Ω in the signal line is always necessary in the case of a low-impedance power supply.

The voltage drop to be taken into account is:
Max. 6 V for 250 Ω communication resistor

HART device block diagram, connection with RIA15

FMR20 with RIA15 (incl. option for FMR20 basic configuration)

The RIA15 remote indicator can be ordered together with the device.

Product structure, feature 620 "Accessory enclosed":
- Option R4 "Remote indicator RIA15 non-hazardous area, field housing"
- Option R5 "Remote indicator RIA15 with explosion protection approval, field housing"

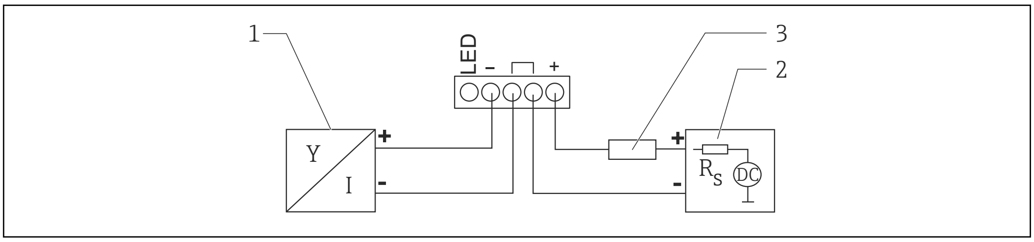
Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

Terminal assignment RIA15
- **+**
  Positive connection, current measurement
- **-**
  Negative connection, current measurement (without backlighting)
- **LED**
  Negative connection, current measurement (with backlighting)
- ⏚
  Functional grounding: Terminal in housing

The RIA15 process indicator is loop-powered and does not require any external power supply.

The voltage drop to be taken into account is:
- ≤1 V in the standard version with 4 to 20 mA communication
- ≤1.9 V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART device and RIA15 without backlighting



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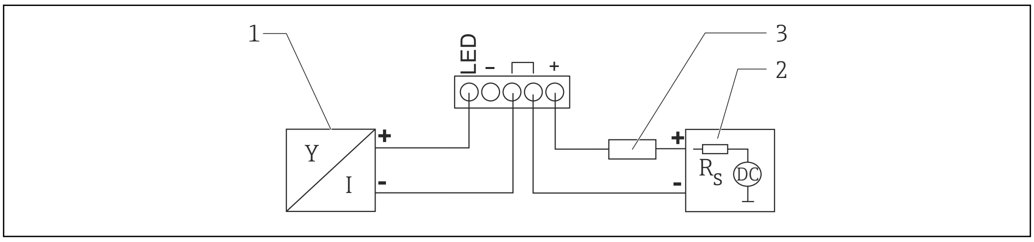
5 *Block diagram of HART device with RIA15 process indicator without light*

1 *Device with HART communication*
2 *Power supply*
3 *HART resistor*

Connection of the HART device and RIA15 with backlighting



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6 *Block diagram of HART device with RIA15 process indicator with light*

1 *Device with HART communication*
2 *Power supply*
3 *HART resistor*

Block diagram of HART device, RIA15 with installed HART communication resistor module

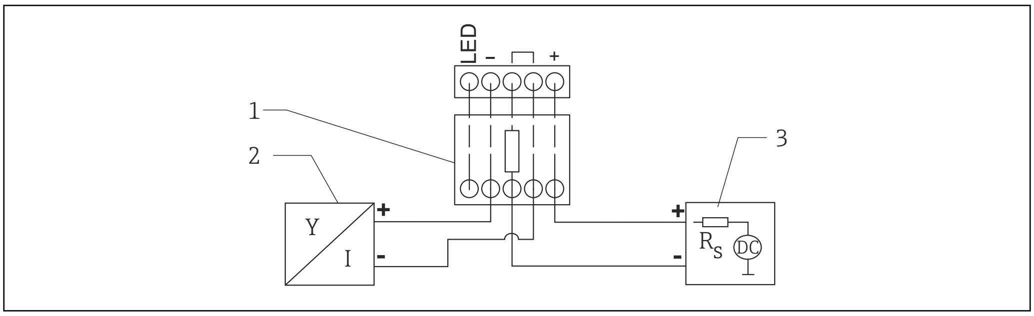
The HART communication module for installation in the RIA15 can be ordered together with the device.

Product structure, feature 620 "Accessory enclosed":
Option R6 "HART communication resistor hazardous / non-hazardous area"

The voltage drop to be taken into account is:
Max. 7 V

Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

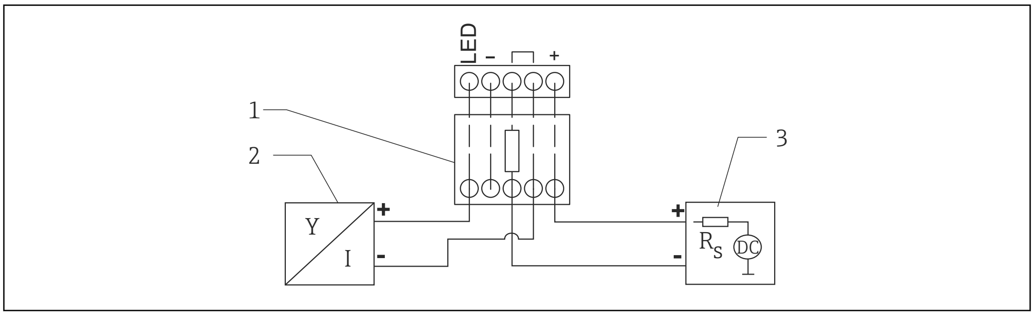
Connection of the HART communication resistor module, RIA15 without backlighting



7 *Block diagram of HART device, RIA15 without light, HART communication resistor module*

1 *HART communication resistor module*
2 *Device with HART communication*
3 *Power supply*

Connection of the HART communication resistor module, RIA15 with backlighting



8 *Block diagram of HART device, RIA15 with light, HART communication resistor module*

1 *HART communication resistor module*
2 *Device with HART communication*
3 *Power supply*

Cable specification

Unshielded cable, wire cross-section 0.75 mm^2
- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC/EN 60079-11 section 10.9, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

The device is supplied with 5 m (16 ft) cable length as standard. Cable lengths 10 m (33 ft) and 20 m (66 ft) are optionally available.

Lengths can be selected by the user up to an overall length of 300 m (980 ft) and are available by the meter (order option "8") or foot (order option "A").

Overvoltage protection

The device is equipped with integrated overvoltage protection.

Performance characteristics

Reference operating conditions

- Temperature = +24 °C (+75 °F) ±5 °C (±9 °F)
- Pressure = 960 mbar abs. (14 psia) ±100 mbar (±1.45 psi)
- Humidity = 60 % ±15 %
- Reflector: metal plate with diameter ≥ 1 m (40 in)
- No major interference reflections inside the signal beam

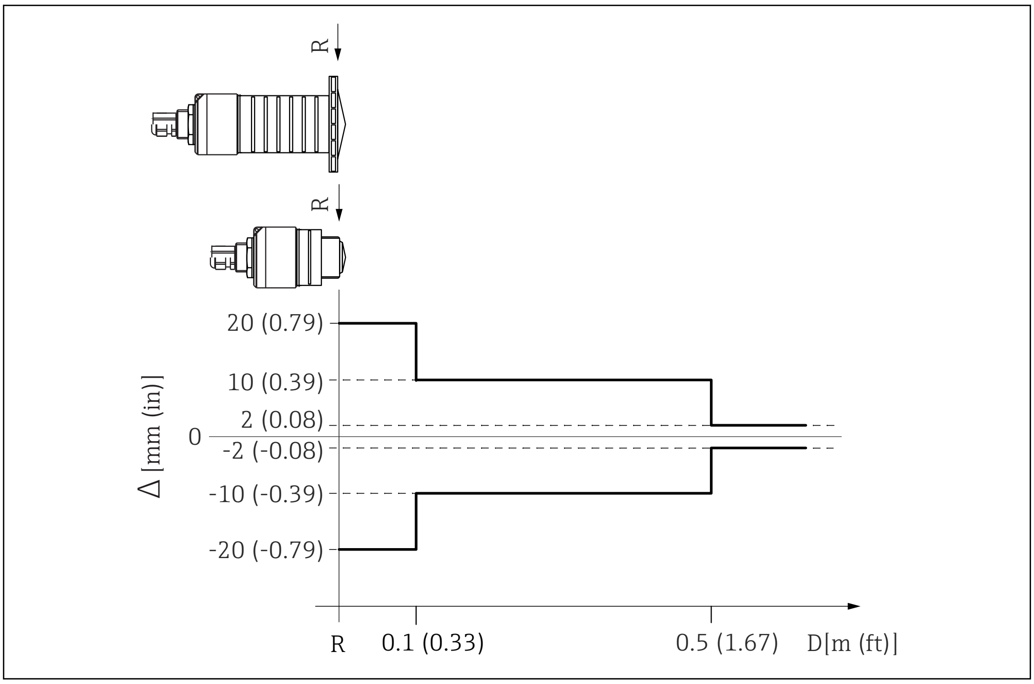
Maximum measured error

Typical data under reference operating conditions: DIN EN 61298-2, percentage values in relation to the span.

Output, digital
(HART, SmartBlue (app))
- Sum of non-linearity, non-repeatability and hysteresis: ±2 mm (±0.08 in)
- Offset/zero point: ±4 mm (±0.16 in)

Output, analog
Only relevant for 4-20mA current output; add error of the analog value to the digital value
- Sum of non-linearity, non-repeatability and hysteresis: ±0.02 %
- Offset/zero point: ±0.03 %

Differing values in near-range applications



9 *Maximum measured error in near-range applications; values for standard version*

Δ Maximum measured error
R Reference point of the distance measurement
D Distance from reference point of antenna

Measured value resolution

Dead band as per EN61298-2:
- Digital: 1 mm (0.04 in)
- Analog: 4 µA

Response time

The response time can be configured. The following step response times apply (in accordance with DIN EN 61298-2) when damping is switched off:

Tank height
<20 m (66 ft)

Sampling rate
1 s^{-1}

Response time

<3 s

In accordance with DIN EN 61298-2, the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90% of the steady-state value for the first time.

Influence of ambient temperature

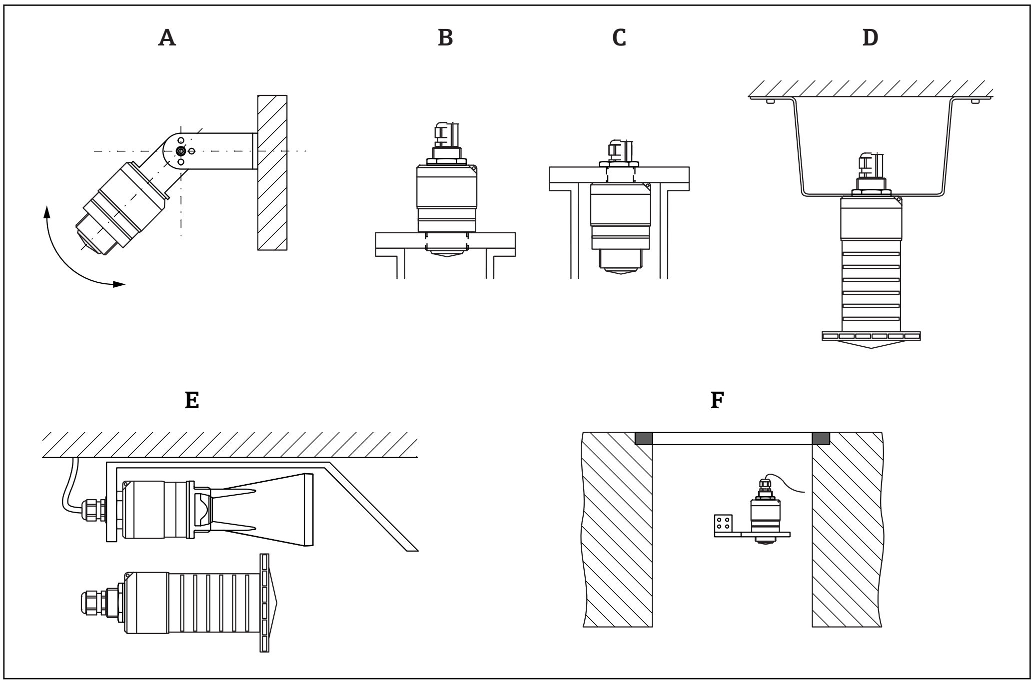
The measurements are carried out in accordance with EN 61298-3

- Digital (HART, *Bluetooth*® wireless technology):
  Standard version: average T_C = ±3 mm (±0.12 in)/10 K
- Analog (current output):
  - Zero point (4 mA): average T_K = 0.02 %/10 K
  - Span (20 mA): average T_K = 0.05 %/10 K

Installation

Installation conditions

Installation types



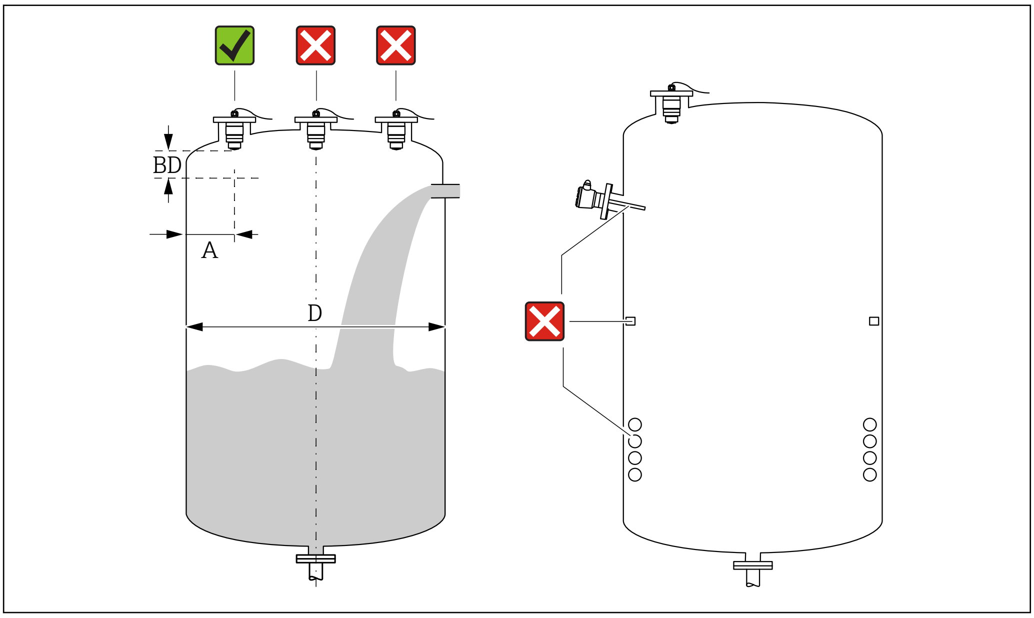
10 Wall, ceiling or nozzle installation

A Wall or ceiling mount, adjustable
B Mounted at front thread
C Mounted at rear thread
D Ceiling installation with counter nut (included in delivery)
E Horizontal installation in cramped spaces (wastewater shaft), 40 mm (1.5 in) antenna with flooding protection tube (accessory) or 80 mm (3 in) antenna recommended
F Shaft wall mounting

Caution!

- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- Always operate the device in a vertical position in free-space applications.

Position for installation on a vessel



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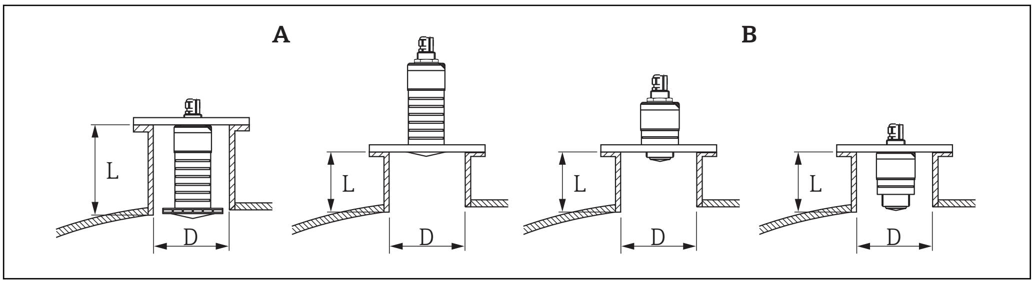
11 *Installation position on a vessel*

- If possible install the sensor so that its lower edge projects into the vessel.
- Recommended distance **A** wall - nozzle outer edge: ~ ⅙ of the vessel diameter **D**. Under no circumstances should the device be mounted closer than 15 cm (5.91 in) to the vessel wall.
- Do not install the sensor in the middle of the vessel.
- Avoid measurements through the filling curtain.
- Avoid equipment such as limit switches, temperature sensors, baffles, heating coils etc.
- No signals are evaluated within the Blocking distance (BD). It can therefore be used to suppress interference signals (e.g. the effects of condensate) in the vicinity of the antenna.
  An automatic Blocking distance of at least 0.1 m (0.33 ft) is configured as standard. However, this can be overwritten manually (0 m (0 ft) is also permitted).
  Automatic calculation:
  Blocking distance = Empty calibration - Full calibration - 0.2 m (0.656 ft).
  Each time a new entry is made in the **Empty calibration** parameter or **Full calibration** parameter, the **Blocking distance** parameter is recalculated automatically using this formula.
  If the result of the calculation is a value < 0.1 m (0.33 ft), the Blocking distance of 0.1 m (0.33 ft) will continue to be used.

Nozzle installation

The antenna should be located out of the nozzle for optimum measurement. The interior of the nozzle must be smooth and may not contain any edges or welded joints. The edge of the nozzle should be rounded if possible.



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12 *Nozzle installation*

A 80 mm (3 in) antenna
B 40 mm (1.5 in) antenna

The maximum length of the nozzle **L** depends on the nozzle diameter **D**.

Please note the limits for the diameter and length of the nozzle.

~~**80 mm (3 in) antenna, installation inside nozzle**~~
- ~~D: min. 120 mm (4.72 in)~~
- ~~L: max. 205 mm (8.07 in) + D × 4.5~~

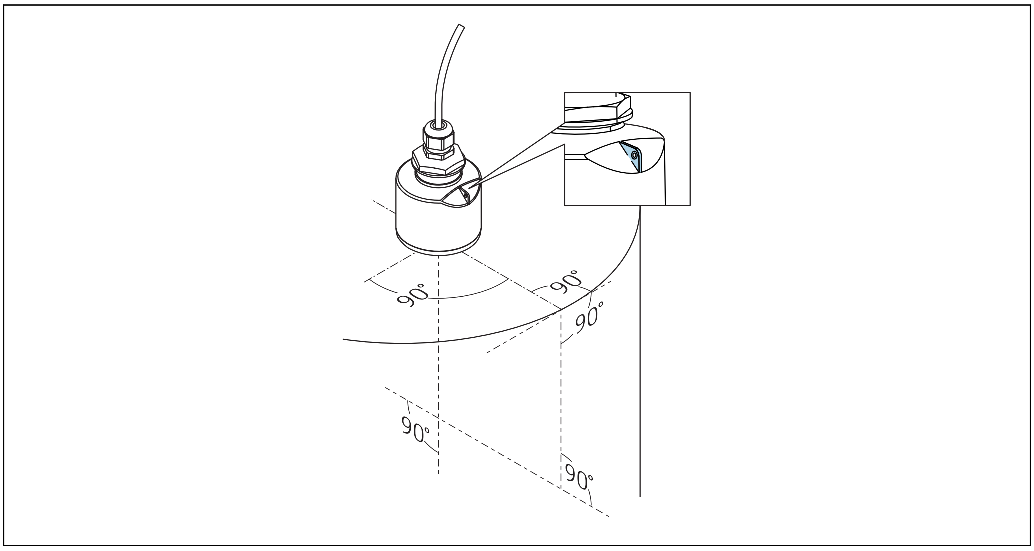
~~**80 mm (3 in) antenna, installation outside nozzle**~~
- ~~D: min. 80 mm (3 in)~~
- ~~L: max. D × 4.5~~

40 mm (1.5 in) antenna, installation outside nozzle
- D: min. 40 mm (1.5 in)
- L: max. D × 1.5

~~**40 mm (1.5 in) antenna, installation inside nozzle**~~
- ~~D: min. 80 mm (3 in)~~
- ~~L: max. 140 mm (5.5 in) + D × 1.5~~

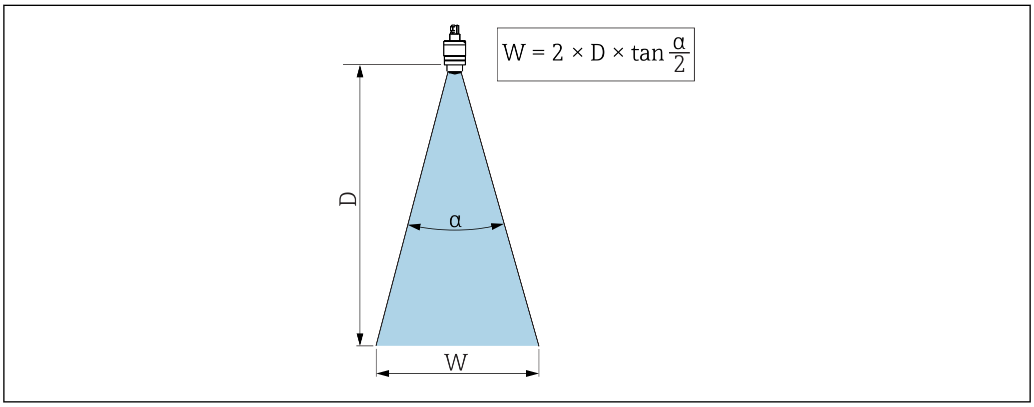
Device alignment for installation on a vessel

- Align the antenna vertically to the product surface.
- Align the eyelet with lug towards the vessel wall as well as possible.



13 *Device alignment for installation on a vessel*

Beam angle



14 *Relationship between beam angle α, distance D and beamwidth diameter W*

The beam angle is defined as the angle **α**, at which the power density of the radar waves reaches half the value of the maximum power density (3 dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.

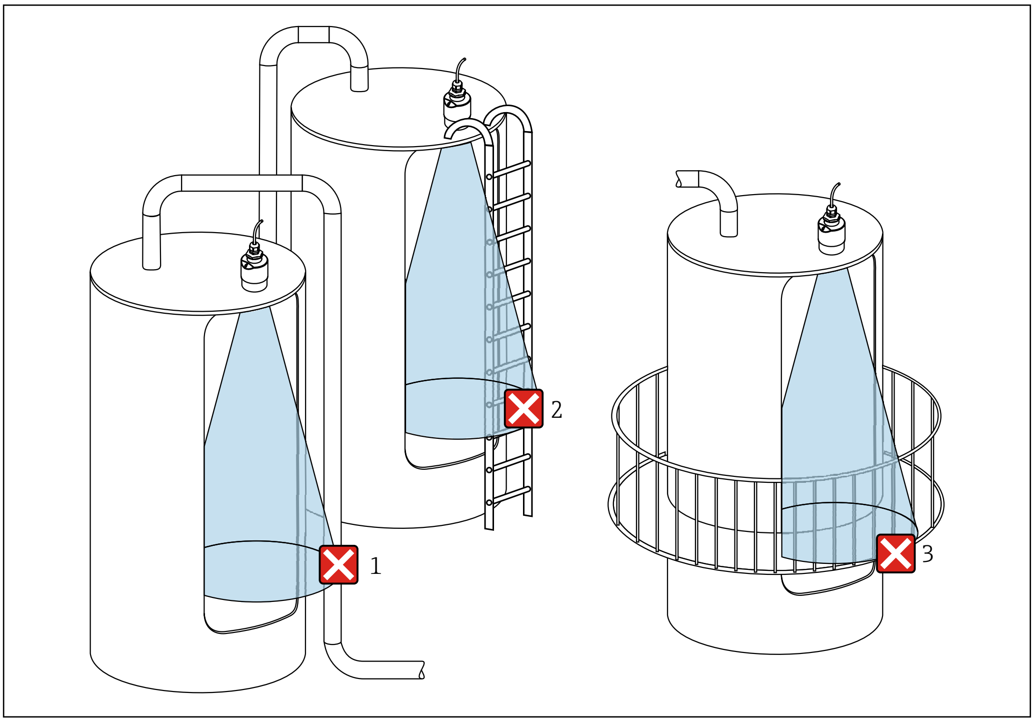
Beam diameter **W** as a function of beam angle **α** and distance **D**.

40 mm (1.5 in) antenna, α 30 °
W = D × 0.54

~~**40 mm (1.5 in) antenna with flooding protection tube, α 12 °**~~
~~W = D × 0.21~~

~~**80 mm (3 in) antenna with or without flooding protection tube, α 12 °**~~
~~W = D × 0.21~~

Measurement in plastic vessels



15 *Measurement in a plastic vessel with a metallic, interfering installation outside of the vessel*

1 *Pipe, tubing*
2 *Ladder*
3 *Grate, railing*

If the outer wall of the vessel is made of a non-conductive material (e.g. GFR), microwaves can also be reflected by interfering installations outside of the vessel.

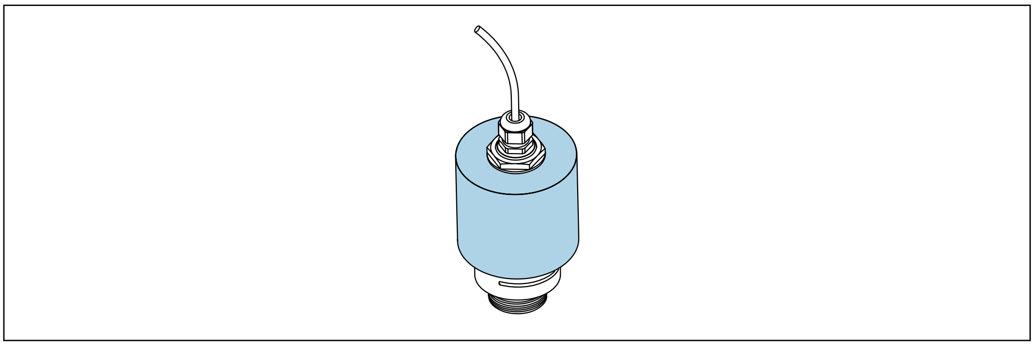
Please ensure there are no interfering installations made of a conductive material in the signal beam (see the beam angle section for information on calculating the beamwidth diameter).

Please contact the manufacturer for further information.

Protective hood

For outdoor use, a protective hood is recommended.

The protective hood can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



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16 *Protective hood, e.g. with 40 mm (1.5") antenna*

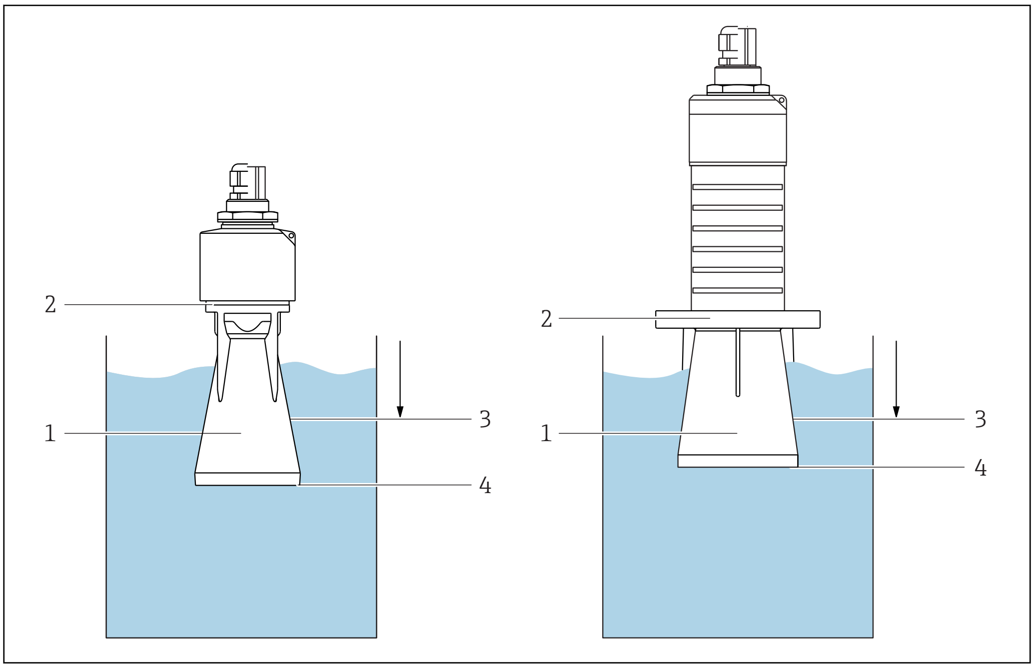
The sensor is not completely covered by the protective hood.

Free-field measurement with flooding protection tube

The flooding protection tube ensures the sensor measures the maximum level even if it is completely flooded.

In free-field installations and/or in applications where there is a risk of flooding, the flooding protection tube must be used.

The flooding protection tube can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



A0031093

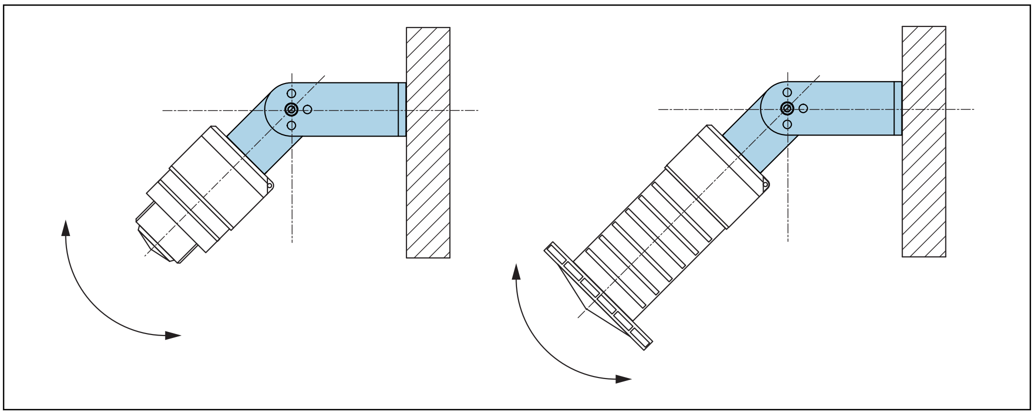
17 *Function of flooding protection tube*

1 *Air pocket*
2 *O-ring (EPDM) seal*
3 *Blocking distance*
4 *Max. level*

The tube is screwed directly onto the sensor and seals off the system by means of an O-ring making it air-tight. In the event of flooding, the air pocket that formed in the tube ensures the measurement of the maximum level at the end of the tube. Due to the fact that the Blocking distance is inside the tube, multiple echoes are not analyzed.

Installation with mounting bracket, adjustable

The mounting bracket can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



18 *Installation with mounting bracket, adjustable*

- Wall or ceiling installation is possible.
- Using the mounting bracket, position the antenna so that it is perpendicular to the product surface.

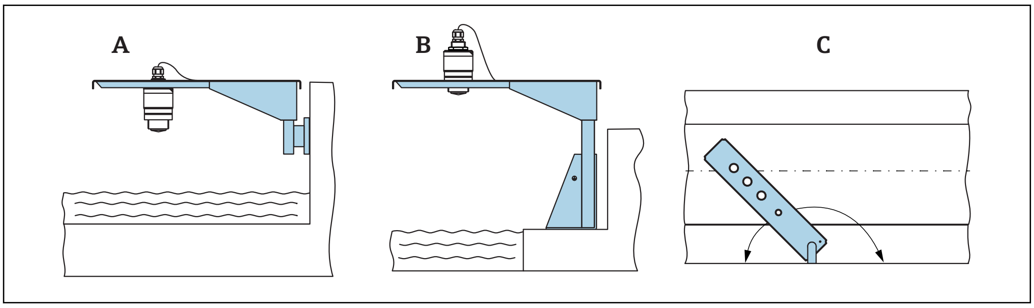
NOTICE

There is no conductive connection between the mounting bracket and transmitter housing.
Electrostatic charging possible.

▶ Integrate the mounting bracket in the local potential equalization system.

Cantilever installation, with pivot

The cantilever, wall bracket and mounting frame are available as accessories.

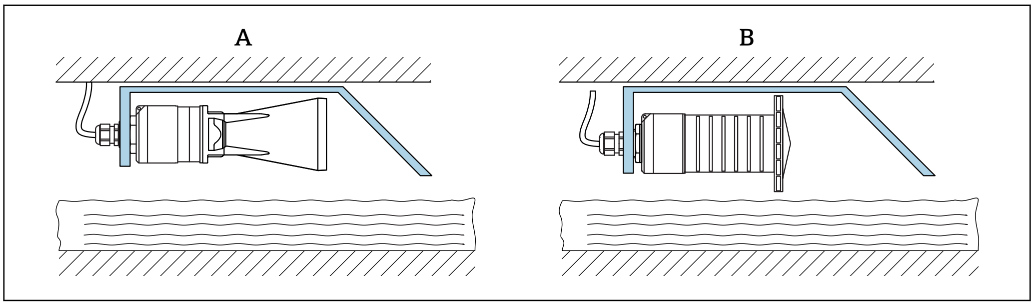


19 *Cantilever installation, with pivot*

A *Cantilever with wall bracket*
B *Cantilever with mounting frame*
C *Cantilever can be turned (e.g., in order to position the device over the center of the flume)*

Installation of horizontal mounting bracket for sewer shafts

The horizontal mounting bracket for sewer shafts is available as an accessory.



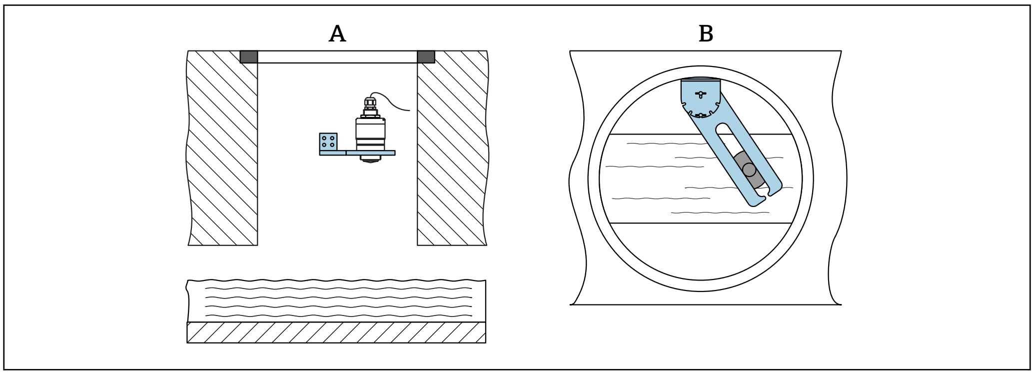
A0043467

20 *Installation of horizontal mounting bracket for sewer shafts*

A 40 mm (1.5 in) Antenna, with flooding protection tube (accessory)
B 80 mm (3 in) Antenna, without flooding protection tube

Mounting in a shaft

The pivoted mounting bracket is available as an accessory.



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21 *Mounting in a shaft, pivotable and adjustable*

A Arm with wall bracket
B Pivotable and adjustable arm (e.g. to align the device with the center of a channel)

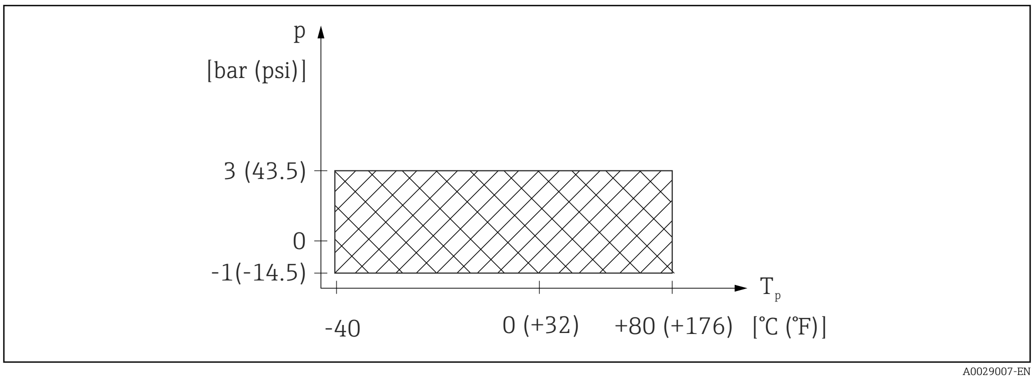
Environment

Ambient temperature range	Measuring device: −40 to +80 °C (−40 to +176 °F) It may not be possible to use the Bluetooth connection at ambient temperatures > 60 °C (140 °F). Outdoor operation in strong sunlight: ▪ Mount the device in the shade. ▪ Avoid direct sunlight, particularly in warm climatic regions. ▪ Use a weather protection cover.
Storage temperature	−40 to +80 °C (−40 to +176 °F)
Climate class	DIN EN 60068-2-38 (test Z/AD)
Operating altitude as per IEC 61010-1 Ed.3	Generally up to 2 000 m (6 600 ft) above sea level.
Degree of protection	Tested acc. to: ▪ IP66, NEMA 4X ▪ IP68, NEMA 6P (24 h at 1.83 m (6.00 ft)1.83 m under water)
Vibration resistance	DIN EN 60068-2-64/IEC 60068-2-64: 20 to 2 000 Hz, 1 $(m/s^2)^2$/Hz

Electromagnetic compatibility (EMC)	Electromagnetic compatibility in accordance with all of the relevant requirements outlined in the EN 61000 series and NAMUR Recommendation EMC (NE 21). Details are provided in the Declaration of Conformity (www.endress.com/downloads).

Process

Process temperature, process pressure



22 FMR20: Permitted range for process temperature and process pressure

Process temperature range
−40 to +80 °C (−40 to +176 °F)

Process pressure range, threaded process connection
- p_{rel} = −1 to 3 bar (−14.5 to 43.5 psi)
- p_{abs} < 4 bar (58 psi)

~~**Process pressure range, UNI flange process connection**~~
- ~~p_{rel} = −1 to 1 bar (−14.5 to 14.5 psi)~~
- ~~p_{abs} < 2 bar (29 psi)~~

The pressure range may be further restricted in the event of a CRN approval.

Dielectric constant

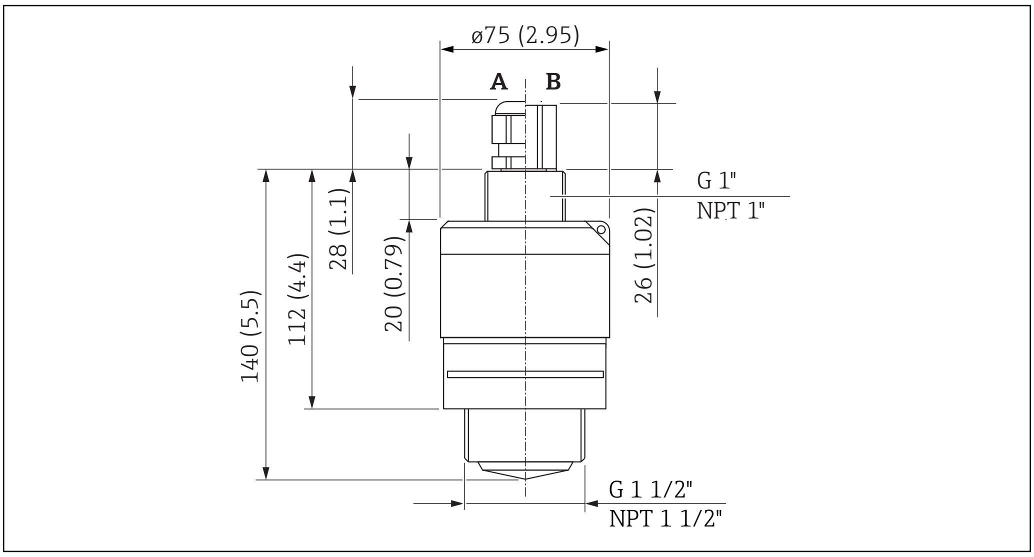
For liquids
- $\varepsilon_r \geq 4$
- Contact Endress+Hauser for lower ε_r values

For dielectric constants (DC values) of many media commonly used in various industries refer to:
- the Endress+Hauser DC manual (CP01076F)
- the Endress+Hauser "DC Values App" (available for Android and iOS)

Mechanical construction

Dimensions

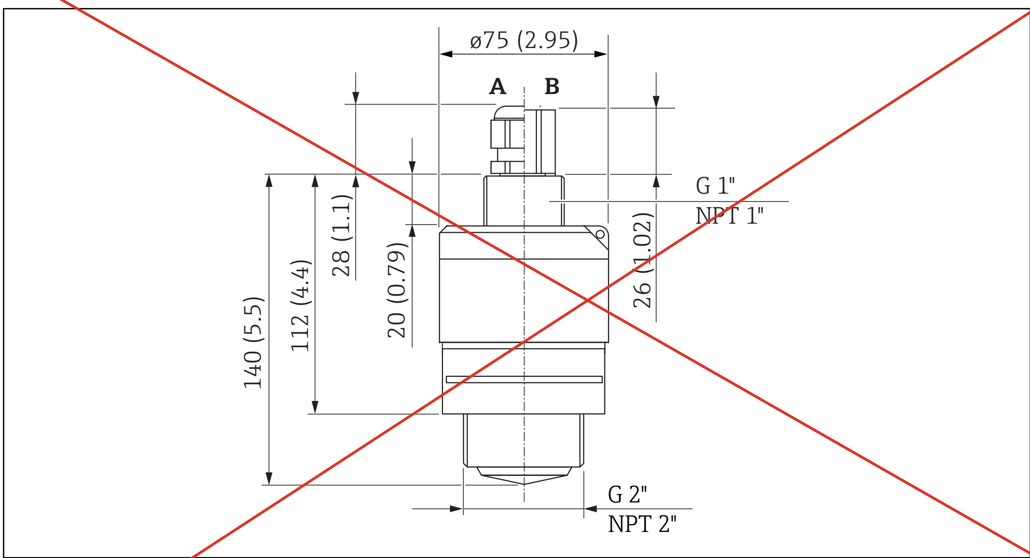
40 mm (1.5 in) Antenna with G 1-½" or MNPT 1-½" thread



23 Dimensions of G 1-½" or MNPT 1-½" process connection thread, engineering unit: mm (in)

A Cable gland
B FNPT ½" conduit

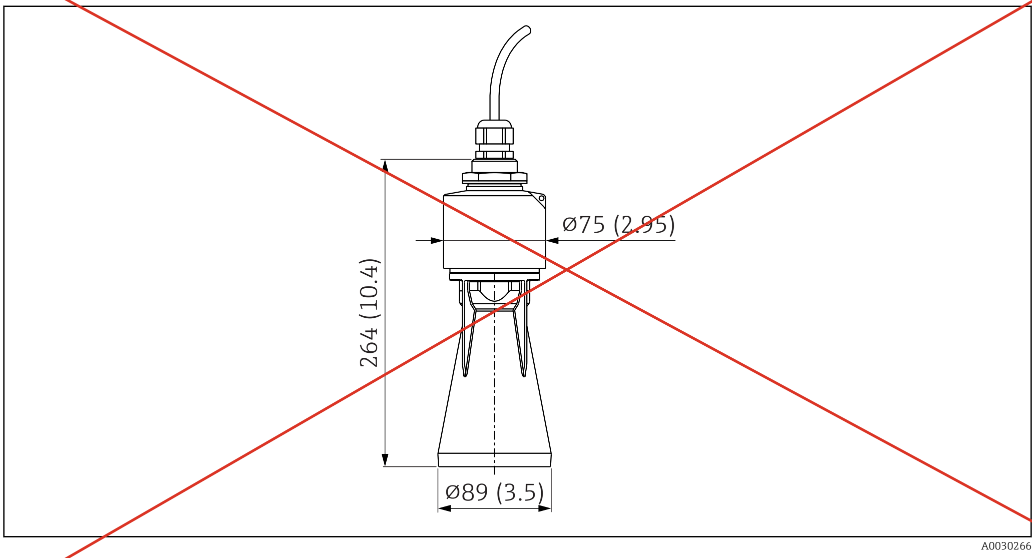
40 mm (1.5 in) Antenna with G 2" or MNPT 2" thread



24 Dimensions of G 2" or MNPT 2" process connection thread, engineering unit: mm (in)

A Cable gland
B FNPT ½" conduit

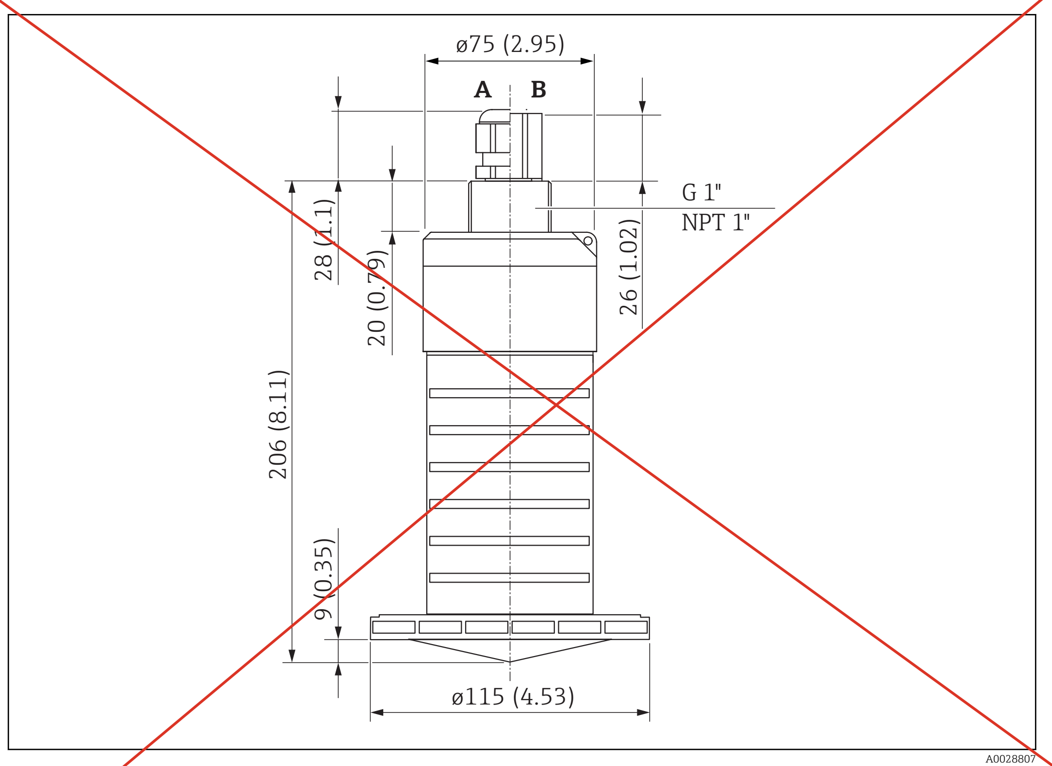
40 mm (1.5 in) antenna with flooding protection tube



25 *Dimensions of 40 mm (1.5 in) antenna mounted with flooding protection tube, engineering unit: mm (in)*

The flooding protection tube, metalized PBT-PC, can be ordered together with the device via the product structure "Accessory enclosed".

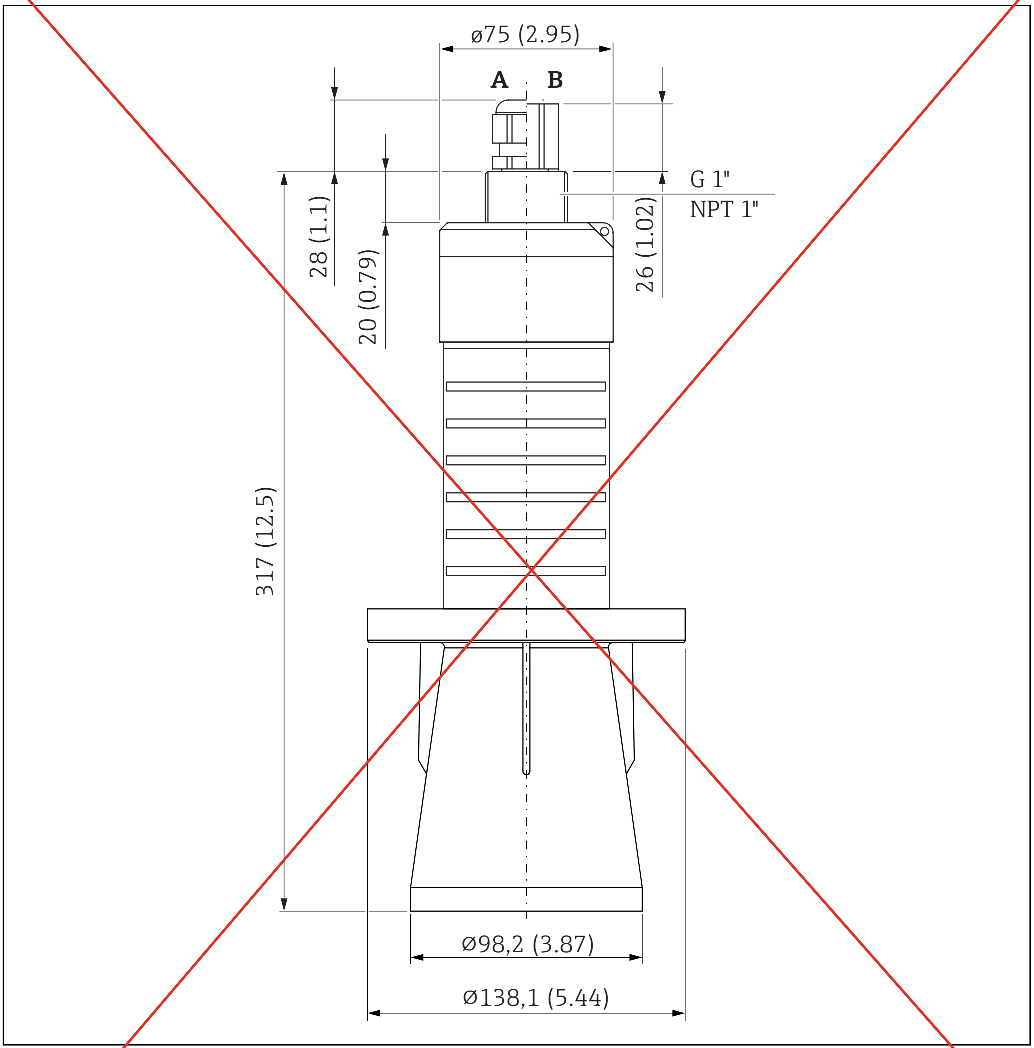
80 mm (3 in)Antenna



26 *Dimensions of 80 mm (3 in) antenna; engineering unit: mm (in)*

A Cable gland
B FNPT ½“ conduit

80 mm (3 in) antenna with flooding protection tube

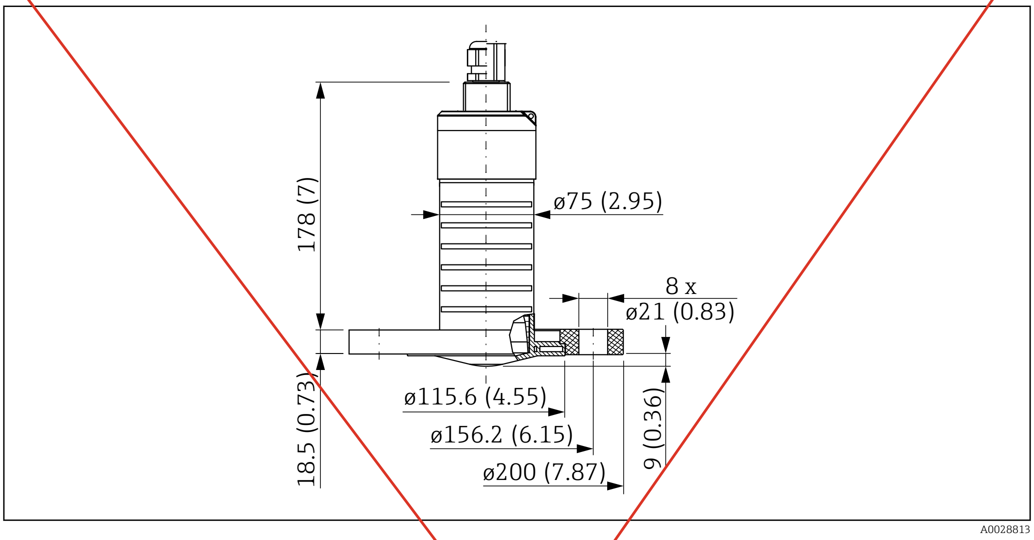


27 *Dimensions of 80 mm (3 in) antenna mounted with flooding protection tube, engineering unit: mm (in)*

A Cable gland
B FNPT ½“ conduit

The flooding protection tube, metalized PBT-PC, can be ordered together with the device via the product structure "Accessory enclosed".

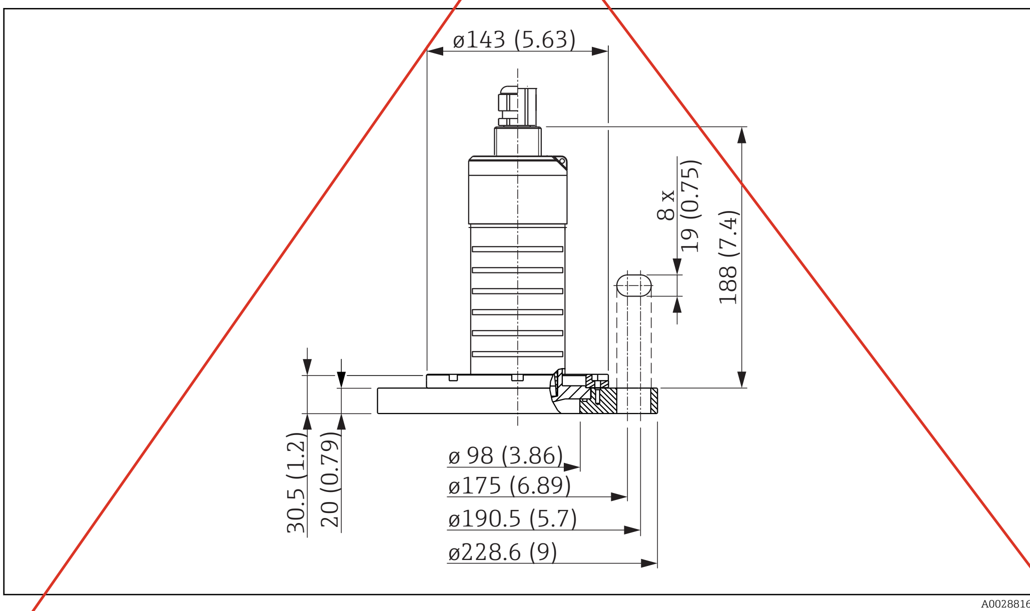
80 mm (3 in) antenna with slip-on flange 3"/DN80



28 *Dimensions of 80 mm (3 in) antenna with slip-on flange 3"/DN80, engineering unit: mm (in)*

The slip-on flange 3"/DN80, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

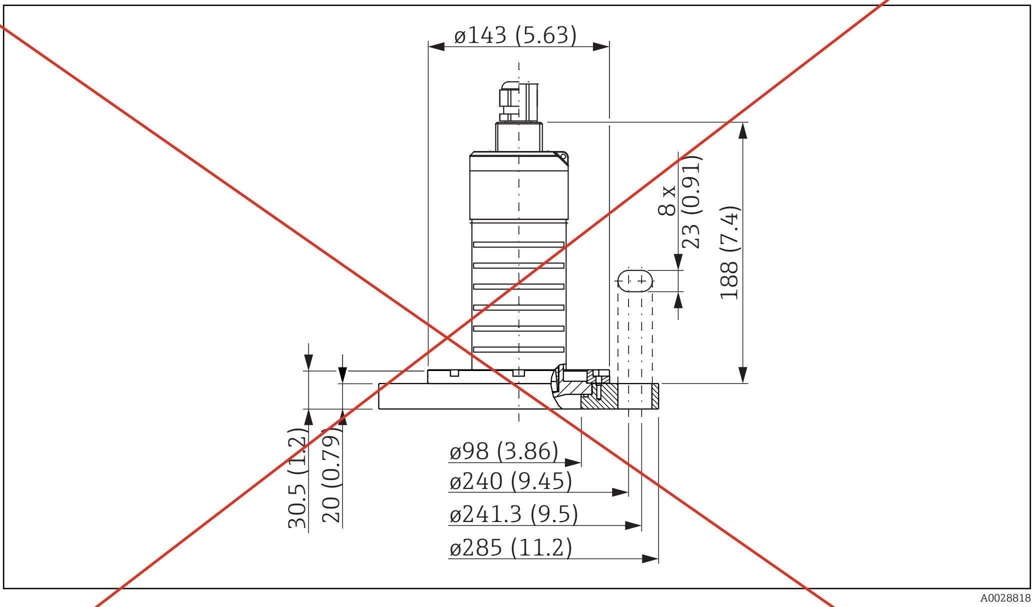
80 mm (3 in) antenna with slip-on flange 4"/DN100



29 *Dimensions of 80 mm (3 in) antenna with slip-on flange 4"/DN100, engineering unit: mm (in)*

The slip-on flange 4"/DN100, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

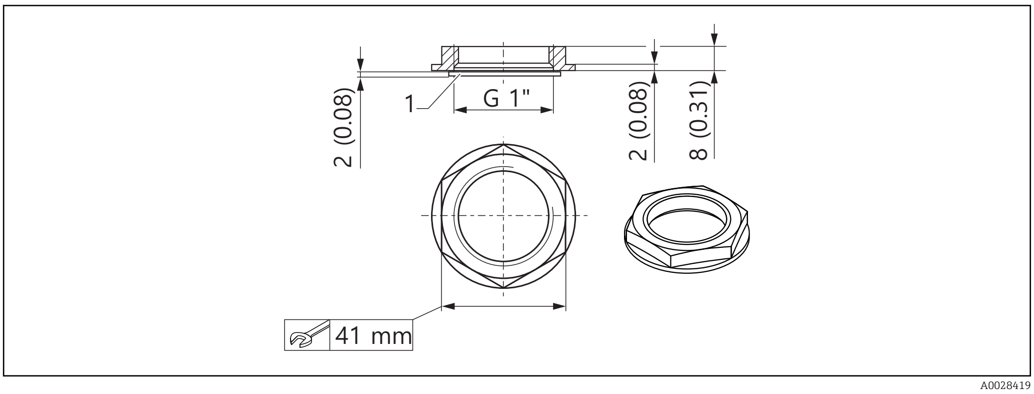
80 mm (3 in) antenna with slip-on flange 6"/DN150



30 Dimensions of 80 mm (3 in) antenna with slip-on flange 6"/DN150, engineering unit: mm (in)

The slip-on flange 6"/DN150, PVDF, can be ordered together with the device via the product structure "Accessory enclosed".

Counter nut for process connection, rear side



31 Dimensions of counter nut for process connection, rear side, engineering unit: mm (in)

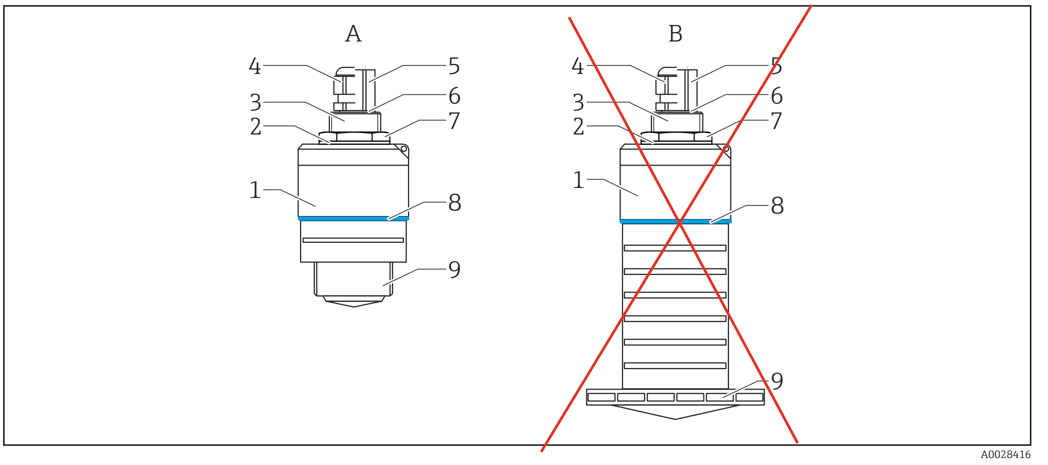
1 Seal

- The counter nut with seal (EPDM) is included in the scope of supply.
- Material: PA66

Weight

Weight (incl. 5 m (16.4 ft) cable)
- Device with 40 mm (1.5 in) antenna: approx. 2.5 kg (5.5 lb)
- Device with 80 mm (3 in) antenna: approx. 2.8 kg (6.2 lb)

Materials



32 *Overview of materials*

A 40 mm (1.5 in)Antenna
~~*B 80 mm (3 in)Antenna*~~
1 Sensor housing; PVDF
2 Seal; EPDM
3 Process connection, rear side; PVDF
4 Cable gland; PA
5 Conduit adapter; CuZn nickel-plated
6 O-ring; EPDM
7 Counter nut; PA6.6
8 Design ring; PBT-PC
9 Process connection, front side; PVDF

Connecting cable

Available cable length: 5 to 300 m (16 to 980 ft)

20m

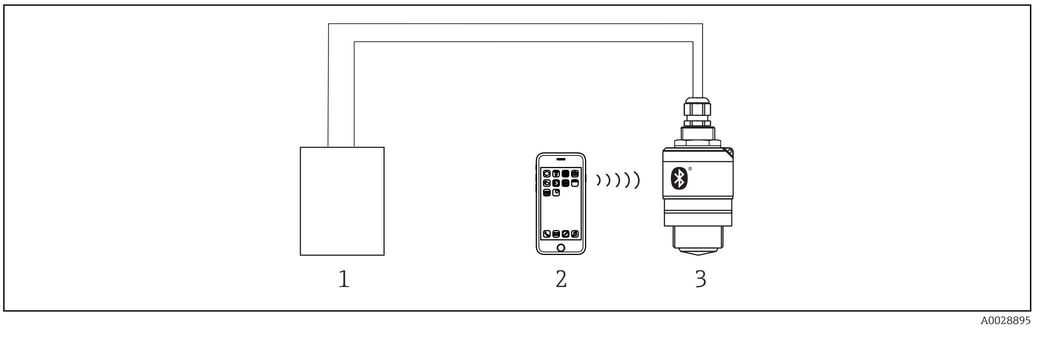
Material : PVC

Operability

Operating concept

- 4 to 20 mA, HART
- Menu guidance with brief explanations of the individual parameter functions in the operating tool
- Optional: SmartBlue (app) via *Bluetooth®* wireless technology

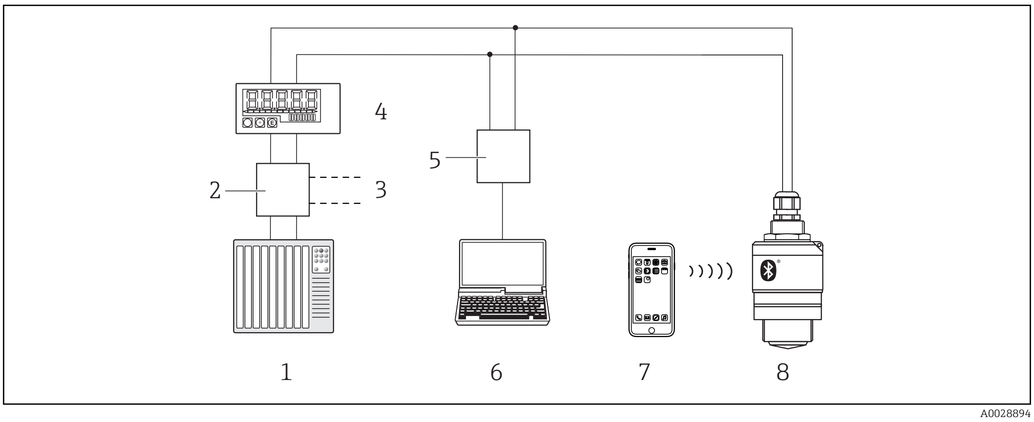
Operation via Bluetooth® wireless technology



33 *Possibilities for remote operation via Bluetooth® wireless technology*

1 Transmitter power supply unit
2 Smartphone/tablet with SmartBlue (app)
3 Transmitter with Bluetooth® wireless technology

Via HART protocol



34 *Options for remote operation via HART protocol*

1 *PLC (programmable logic controller)*
2 *Transmitter power supply unit, e.g. RN221N (with communication resistor)*
3 *Connection for Commubox FXA195*
4 *Loop-powered RIA15 process indicator*
5 *Commubox FXA195 (USB)*
6 *Computer with operating tool (FieldCare, DeviceCare)*
7 *Smartphone / tablet with SmartBlue (app)*
8 *Transmitter with Bluetooth® wireless technology*

Certificates and approvals

The availability of approvals and certificates can be called up daily via the Product Configurator.

CE mark

The measuring system meets the legal requirements of the applicable EU Directives. These are listed in the corresponding EU Declaration of Conformity along with the standards applied.

Endress+Hauser confirms successful testing of the device by affixing to it the CE mark.

RoHS

The measuring system complies with the substance restrictions of the Restriction on Hazardous Substances Directive 2011/65/EU (RoHS 2).

EAC conformity

The measuring system meets the legal requirements of the applicable EAC guidelines. These are listed in the corresponding EAC Declaration of Conformity together with the standards applied.

Endress+Hauser confirms successful testing of the device by affixing to it the EAC mark.

RCM-Tick marking

The supplied product or measuring system meets the ACMA (Australian Communications and Media Authority) requirements for network integrity, interoperability, performance characteristics as well as health and safety regulations. Here, especially the regulatory arrangements for electromagnetic compatibility are met. The products are labelled with the RCM- Tick marking on the name plate.

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Approvals

- Non-hazardous area
- ATEX II 1 G Ex ia IIC T4 Ga
- ATEX II 1/2 G Ex ia IIC T4 Ga/Gb
- CSA C/US General Purpose
- CSA C/US IS Cl.I Div.1 Gr.A-D, AEx ia / Ex ia T4
- CSA C/US Cl.I Div.2 Gr.A-D, T4
- EAC Ex ia IIC T4 Ga/Gb
- IEC Ex ia IIC T4 Ga/Gb
- KC Ex ia IIC T4 Ga/Gb [3)]
- INMETRO Ex ia IIC T4 Ga/Gb
- NEPSI Ex ia IIC T4 Ga/Gb
- TIIS Ex ia IIC T4 [3)]

Additional safety instructions must be followed for use in hazardous areas. Please refer to the separate "Safety Instructions" (XA) document included in the delivery. Reference to the applicable XA can be found on the nameplate.

Explosion-protected smartphones and tablets

Only mobile end devices with Ex approval may be used in hazardous areas.

Pressure equipment with allowable pressure ≤ 200 bar (2 900 psi)

Pressure instruments with a flange and threaded boss that do not have a pressurized housing do not fall within the scope of the Pressure Equipment Directive, irrespective of the maximum allowable pressure.

Reasons:

According to Article 2, point 5 of EU Directive 2014/68/EU, pressure accessories are defined as "devices with an operational function and having pressure-bearing housings".

If a pressure instrument does not have a pressure-bearing housing (no identifiable pressure chamber of its own), there is no pressure accessory present within the meaning of the Directive.

3) Under development at time of going to press

EN 302729-1/2 radio standard

The devices comply with the LPR (Level Probing Radar) radio standard EN 302729-1/2 and are approved for unrestricted use inside and outside of closed vessels in countries of the EU and EFTA. As a prerequisite, the countries in question must have already implemented this standard.

The following countries are those that have currently implemented the standard:

Belgium, Bulgaria, Germany, Denmark, Estonia, France, Greece, UK, Ireland, Iceland, Italy, Liechtenstein, Lithuania, Latvia, Malta, The Netherlands, Norway, Austria, Poland, Portugal, Romania, Sweden, Switzerland, Slovakia, Spain, Czech Republic and Cyprus.

Implementation is still underway in all of the countries not listed.

Please note the following for operation of the devices outside of closed vessels:

1. The device must be mounted in accordance with the instructions in the "Installation" section.
2. Installation must be carried out by properly trained, expert staff.
3. The device antenna must be installed in a fixed location pointing vertically downwards.
4. The installation site must be located at a distance of 4 km from the astronomy stations listed below or otherwise approval must be provided by the relevant authority. If the device is installed at a distance of 4 to 40 km from one of the listed stations, it must not be installed at a height of more than 15 m (49 ft) above the ground.

Astronomy stations

Country	Name of the station	Latitude	Longitude
Germany	Effelsberg	50°31'32" North	06°53'00" East
Finland	Metsähovi	60°13'04" North	24°23'37" East
	Tuorla	60°24'56" North	24°26'31" East
France	Plateau de Bure	44°38'01" North	05°54'26" East
	Floirac	44°50'10" North	00°31'37" West
Great Britain	Cambridge	52°09'59" North	00°02'20" East
	Damhall	53°09'22" North	02°32'03" West
	Jodrell Bank	53°14'10" North	02°18'26" West
	Knockin	52°47'24" North	02°59'45" West
	Pickmere	53°17'18" North	02°26'38" West
Italy	Medicina	44°31'14" North	11°38'49" East
	Noto	36°52'34" North	14°59'21" East
	Sardinia	39°29'50" North	09°14'40" East
Poland	Fort Skala Krakow	50°03'18" North	19°49'36" East
Russia	Dmitrov	56°26'00" North	37°27'00" East
	Kalyazin	57°13'22" North	37°54'01" East
	Pushchino	54°49'00" North	37°40'00" East
	Zelenchukskaya	43°49'53" North	41°35'32" East
Sweden	Onsala	57°23'45" North	11°55'35" East
Switzerland	Bleien	47°20'26" North	08°06'44" East
Spain	Yebes	40°31'27" North	03°05'22" West
	Robledo	40°25'38" North	04°14'57" West
Hungary	Penc	47°47'22" North	19°16'53" East

As a general rule, the requirements outlined in EN 302729-1/2 must be observed.

FCC / Industry Canada

This device complies with Part 15 of the FCC Rules [and with Industry Canada license-exempt RSS standard(s)]. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

[Any] Changes or modifications made to this equipment not expressly approved by Endress+Hauser may void the FCC authorization to operate this equipment.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

- The installation of the LPR/TLPR device shall be done by trained installers, in strict compliance with the manufacturer's instructions.
- The use of this device is on a "no-interference, no-protection" basis. That is, the user shall accept operations of high-powered radar in the same frequency band which may interfere with or damage this device. However, devices found to interfere with primary licensing operations will be required to be removed at the user's expense.
- Only for usage without the accessory "flooding protection tube", i.e. NOT in the free-field: This device shall be installed and operated in a completely enclosed container to prevent RF emissions, which can otherwise interfere with aeronautical navigation.

FCC / Industry Canada IDs

Tank level-probing radar
- **HVIN: FMR20**
  - FCC ID: LCGFMR2XK
  - Industry Canada ID: 2519A-2K
- **HVIN: FMR20X**
  - FCC ID: LCGFMR2XKT
  - Industry Canada ID: 2519A-2KT

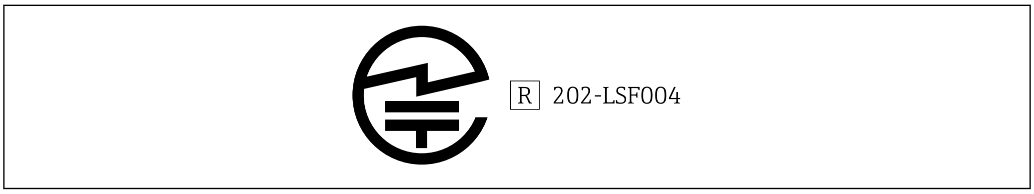
Level-probing radar:
- HVIN: **FMR20+R7; FMR20+R8**
  - FCC ID: LCGFMR2XKF
  - Industry Canada ID: 2519A-2KF
- HVIN: **FMR20+R7X; FMR20+R8X**
  - FCC ID: LCGFMR2XKL
  - Industry Canada ID: 2519A-2KL

Japanese Radio Law and Japanese Telecommunications Business Law Compliance

This device is granted pursuant to the Japanese Radio Law (電波法) and the Japanese Telecommunications Business Law (電気通信事業法). This device should not be modified (otherwise the granted designation number will become invalid).

Certified No.: 202-LSF004

The products are labelled with the Technical Conformity Mark (GITEKI) from Japanese Ministry of Internal Affairs and Communications (MIC) on the name plate.



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Mexico

El funcionamiento de este equipo está sujeto a las dos condiciones siguientes:
- (1) Este equipo o aparato no puede causar interferencias perjudiciales.
- (2) Este equipo o aparato debe aceptar todas las interferencias, incluyendo las que puedan causar un funcionamiento indeseado del equipo o aparato.

Este producto contiene un módulo inalámbrico

Marca: Endress+Hauser

Modelo: FMR20

A0034100

Other standards and guidelines

- IEC/EN 61010-1
  Protection Measures for Electrical Equipment for Measurement, Control, Regulation and Laboratory Procedures
- IEC/EN 55011
  "EMC Emission, RF Emission for Class B". Industrial, scientific and medical equipment – Electromagnetic disturbance characteristics - Limits and methods of measurement
- IEC/EN 61000-4-2
  EMC Immunity, ESD (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrostatic discharge immunity test (ESD)
- IEC/EN 61000-4-3
  EMC Immunity, RF field susceptibility (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Radiated, radio-frequency, electromagnetic field immunity test
- IEC/EN 61000-4-4
  EMC Immunity, bursts (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Electrical fast transient/burst immunity test
- IEC/EN 61000-4-5
  EMC Immunity, surge (Performance Criteria B). Electromagnetic compatibility (EMC): Testing and measurement techniques - Surge immunity test
- IEC/EN 61000-4-6
  EMC Immunity, conducted RF (Performance Criteria A). Electromagnetic compatibility (EMC): Testing and measurement techniques - Immunity to conducted disturbances induced by radio-frequency fields
- IEC/EN 61000-4-8
  EMC Immunity, magnetic fields 50 Hz. Electromagnetic compatibility (EMC): Testing and measurement techniques - Power frequency magnetic field immunity test
- EN 61000-6-3
  EMC Emission, conducted RF. EMC: Radiated interference - Residential, commercial and light industry environment
- NAMUR NE 21
  Electromagnetic compatibility (EMC) of industrial process and laboratory control equipment
- NAMUR NE 43
  Standardization of the signal level for the breakdown information of digital transmitters with analog output signal.
- NAMUR NE 107
  Status classification as per NE107
- NAMUR NE 131
  Requirements for field devices for standard applications
- IEEE 802.15.1
  Requirements for the *Bluetooth*® wireless technology interface

Ordering information

Detailed ordering information is available for your nearest sales organization www.addresses.endress.com or in the Product Configurator under www.endress.com :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.

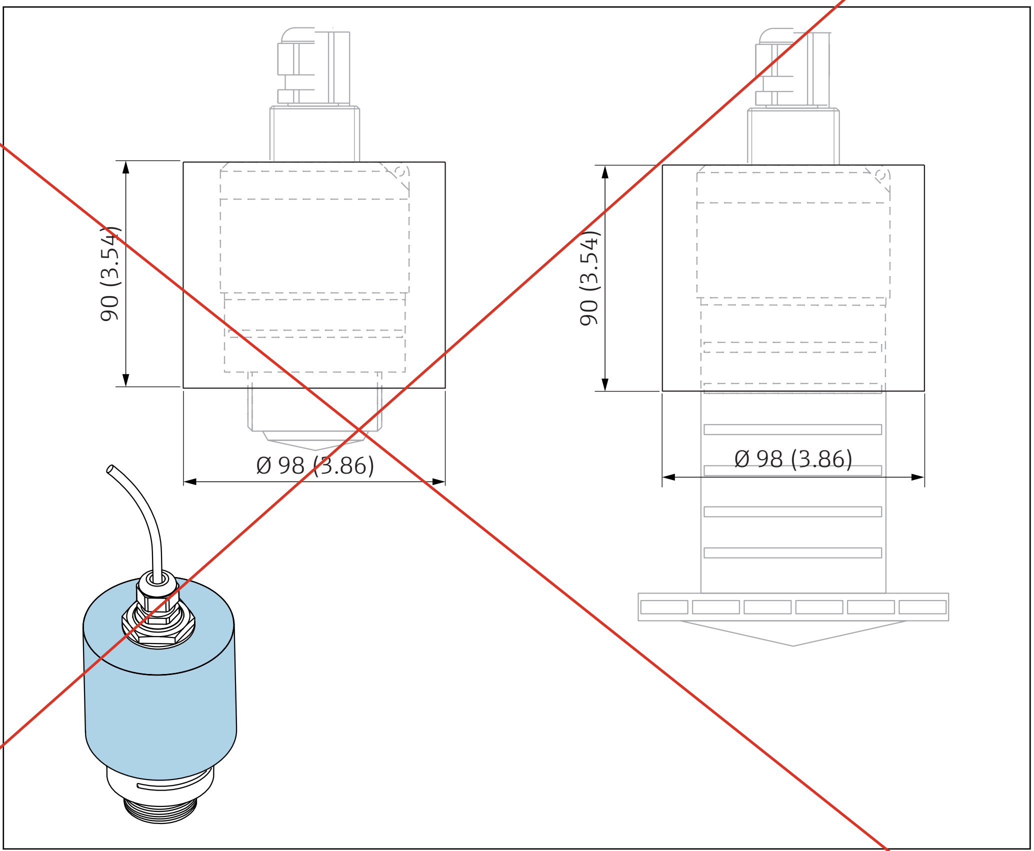
Product Configurator - the tool for individual product configuration
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Device-specific accessories

Protective cover

The protective cover can be ordered together with the device via the product structure "Accessory enclosed".



35 *Dimensions of protective cover, engineering unit: mm (in)*

Material
PVDF

Data Sheet 2

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 01/27/2026
<u>Qty</u>	<u>Description</u>
6	Mfg: Anchor Scientific: Float Switch, NO Model Number: P60NO-SSTNM Tags/Service: LSLL-01, LSL-01, LSL-02, LSH-03, LSHH-01, LSH-02 / Pump Station Level Float Switches Specifications: • P - Pipe mounted for 1″ pipe • 60 - 60ft cable length • NO - Normally open switch configuration • SST - Type 316 stainless steel housing • NM - Non-mercury, direct acting float switch

FLOAT SWITCH, NO

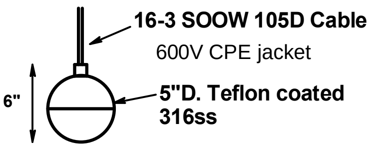
GENERAL DESCRIPTION

TAGS: LSLL-01, LSL-01, LSL-02, LSH-03, LSHH-01, LSH-02

The Roto-Float SSTNM is a non-mercury, direct acting float switch. Each float contains a single pole, single throw reed switch which changes state at +30 from horizontal and -30 from horizontal. The float housing is 5" in diameter, made of 316 stainless steel and Teflon coated for preventing grease and debris from collecting on the float. Extra-hard usage electrical cable is used to electrically connect the switch back to the control panel. The cable and switch are epoxy encapsulated in the float housing potting tube, forming a watertight, impact resistant unit. The Roto-Float SSTNM **Type P** is equipped with a stainless steel bracket for attachment to a 1" pipe, (by others). ~~The **Type W** mounting style is equipped with a stainless steel bracket and clip assembly that secures the float to a WRW wire rope kit. The **Type S** suspended has an external lead weight attached to the cable and is suspended fro above on the floats own cable.~~

Applications

The Roto-Float SSTNM was designed primarily for controlling pumps in wastewater applications for turning pumps on and off in emptying or filling situations; signal alarm levels, and other application where liquid levels need to be automatically monitored. Its use may be suitable in other liquids, as well.



Specification Information

Electrical Rating: 1A @ 120Vac; reed switch
Temp. Limit: 110F
Circuit Configuration: NO, *normally open
~~**NC, *normally closed**~~

This switch is compatible with intrinsically safe circuits. The switch neither stores nor generates an electrical charge.

Mounting Styles:
Type P- Pipe mounted for 1" pipe
~~**Type W- Wire rope mounted, (use w. WRW Kit)**~~
~~**Type S- Suspended, external weight on cable**~~

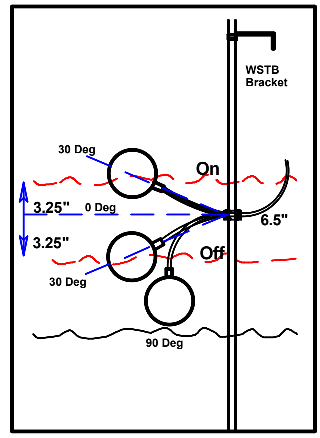
60' cable length

P60NO-SSTNM

Mounting Style	Cable Length	*Switch Configuration	*Model No.	Wgt
~~Type P~~	~~20~~	~~NO~~	~~P20NO-SSTNM~~	~~3.4~~
~~Type P~~	~~30~~	~~NO~~	~~P30NO-SSTNM~~	~~4.3~~
Type P	40	NO	P40NO-SSTNM	5.2
~~Type W~~	~~20~~	~~NO~~	~~W20NO-SSTNM~~	~~3.2~~
~~Type W~~	~~30~~	~~NO~~	~~W30NO-SSTNM~~	~~4.1~~
~~Type W~~	~~40~~	~~NO~~	~~W40NO-SSTNM~~	~~5.0~~
~~Type S~~	~~20~~	~~NO~~	~~S20NO-SSTNM~~	~~7.2~~
~~Type S~~	~~30~~	~~NO~~	~~S30NO-SSTNM~~	~~8.1~~
~~Type S~~	~~40~~	~~NO~~	~~S40NO-SSTNM~~	~~9.0~~

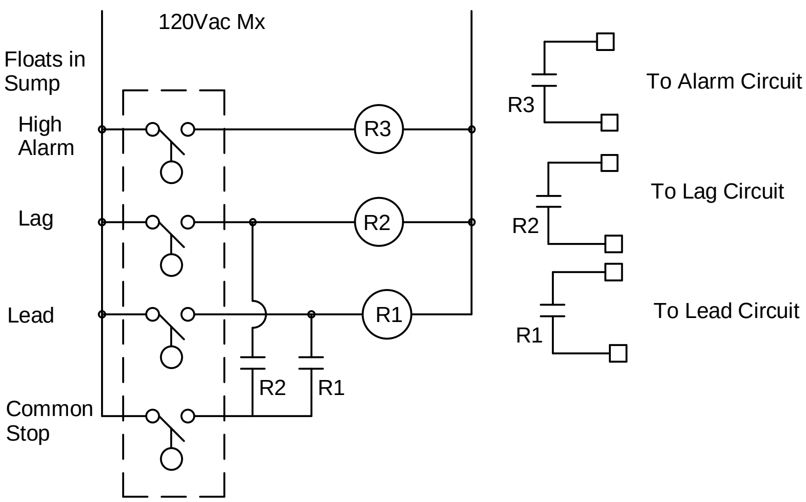
*Normal is defined as state of switch with no liquid present.

*Change 'NO' to 'NC' for normally closed floats. Longer lengths are available.



Normally Open Type P SSTNM
Switch Status vs. Liquid Level

Nominal Duplex Circuit (Reference Only!)



Submitted_____________

Approved_____________

Important Information:

The green wire should always be properly grounded, per NEC. Use this product in accordance with the NEC, local codes, and the authority having jurisdiction. Roto-Float SSTNM is suitable for use with intrinsically safe circuits.

Do not use Roto-Float SSTNM in gasoline or other volatiles and combustibles. Check with the factory for exotic applications.

anchor scientific, inc.
Box 378
Long Lake, MN 55356 Ph: 952-473-7115

Drwn j.p.	**Date** 3/11/11	**Roto-Float SSTNM**
		Rev.
		DWG. SSTNM-BS1

Data Sheet 3

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 01/27/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Endress + Hauser: Digital Chart Recorder Model Number: RSG35-B2A+C1 Tags/Service: RECORDER / PLC/RTU Panel Transmitter Digital Chart Recorder Specifications: • RSG35 - Ecograph T data manager • B - 4 universal inputs • 2 - 24 −10%…+15% AC/DC power supply • A - Ethernet RJ45 + USB communication • C1 - SD card, industrial grade, 1GB
1	Mfg: Endress + Hauser: 1GB Neutral SD Memorycard Model Number: 71187780 Tags/Service: RECORDER / PLC/RTU Panel Transmitter Digital Chart Recorder Specifications: • SD memory card • 1GB storage capacity • Generic, unbranded SD card

TI01079R/09/EN/05.18
71400825
2018-06-29

DIGITAL CHART RECORDER

TAGS: RECORDER

Technical Information
Ecograph T, RSG35

Universal Data Manager

Record, visualize and monitor

Application

The Ecograph T graphic display recorder records and visualizes relevant process values via analog or digital input signals. The measured values are securely saved and limit values are monitored. Furthermore the Ecograph T offers intuitive operation and simple system integration. Remote configuration and visualization of the current and recorded data is easy thanks to the integrated web server - no additional software needs to be installed. In addition the Essential Version of the Field Data Manager software is also supplied with the product as standard. This software can be used to export the recorded data, save the data to an SQL database in a way that the data cannot be manipulated, and visualize the data externally.

The Ecograph T is the right solution for a wide range of applications such as:
- Quality and quantity monitoring in the water and wastewater industry
- Monitoring of processes in power stations
- Displaying and recording of critical process parameters
- Tank and level monitoring
- Temperature monitoring in metal working

Your benefits

- Versatile: up to 12 universal inputs record a wide range of measuring signals
- Clear layout: 5.7" TFT screen for displaying measured values in a maximum of four groups, with digital, bar graph and curve display
- Fast: 100 ms scan rate for all channels
- Compact: low installation depth, saves space and money
- Easy: intuitive operation via the navigator (jog/shuttle dial) on site, or user-friendly operation at the PC via the integrated web server
- Safe: reliable data archiving with internal memory and separate SD card
- Informative: e-mail notification in event of alarms and limit violation
- System-enabled: common interfaces such as Ethernet, RS232/485, USB and optional slave function for Modbus RTU/TCP
- Smart: optional mathematics functions to calculate other values
- WebDAV: data saved on SD card transmitted directly to a PC via HTTP without any additional software.

Endress+Hauser
People for Process Automation

Function and system design

Measuring principle

Electronic acquisition, display, recording, analysis, remote transmission and archiving of analog and digital input signals.

The device is intended for installation in a panel or cabinet. There is also the option of operating it in a desktop housing or field housing.

Measuring system

Multichannel data recording system with multicolored TFT display (145 mm / 5.7" screen size), galvanically isolated universal inputs (U, I, TC, RTD, pulse, frequency), digital inputs, transmitter power supply, limit relay, communication interfaces (USB, Ethernet, optional RS232/485), optionally available with Modbus protocol, 128 MB internal memory, external SD card and USB stick. An Essential Version of the Field Data Manager software is included for SQL-supported data analysis at the PC.

The number of inputs available in the basic device can be individually increased using a maximum of 3 plug-in cards. The device supplies power directly to connected two-wire transmitters. The device is configured and operated via the navigator (jog/shuttle dial), via the integrated web server and a PC, or via an external keyboard. Online help facilitates local operation.

Reliability

Dependability

Depending on the device version, the MTBF is between 52 years and 24 years (calculated based on SN29500 standard at 40°C)

Serviceability

Battery-backed time and data memory. It is advisable to have the backup battery replaced by a service technician after 10 years.

Real time clock (RTC)

- Configurable summer/normal time changeover
- Battery buffer. It is advisable to have the backup battery replaced by a service technician after 10 years.
- Drift: <10 min./year
- Time synchronization possible via SNTP or via digital input.

Standard diagnostic functions as per Namur NE 107

The diagnostic code is made up of the error category as per Namur NE 107 and the message number.

- Cable open circuit, short-circuit
- Incorrect wiring
- Internal device errors
- Overrange/underrange detection
- Ambient temperature out-of-range detection

Device error/alarm relay

One relay can be used as an alarm relay. If the device detects a system error (e.g. hardware defect) or a malfunction (e.g. cable open circuit), the selected output/relay switches.

This "alarm relay" switches if "F"-type errors occur (F = failure), i.e. "M"-type errors (M= Maintenance required) do not switch the alarm relay.

Safety

The tamper-proof recorded data are saved and can be transferred to an external SQL database for archiving in a way that prevents subsequent manipulation.

IT security

The manufacturer only provides a warranty if the device is installed and used as described in the Operating Instructions. The device is equipped with security mechanisms to protect it against any inadvertent changes to the device settings.

IT security measures in line with operators' security standards and designed to provide additional protection for the device and device data transfer must be implemented by the operators themselves.

Input

Measured values

Number of analog universal inputs

~~Standard version without universal inputs~~. Optional input cards (slot 1-3) with 4 universal inputs (4/~~8/12~~) each.

Number of digital inputs

6 digital inputs

Number of mathematics channels

4 mathematics channels (optional). Mathematics functions can be freely edited via a formula editor.

Integration of calculated values e.g. for totalization.

Number of limit values

30 limit values (individual channel assignment)

Function of analog universal inputs

You are free to choose between the following measured variables for each universal input: U, I, RTD, TC, pulse input or frequency input.

Integration of input variable for totalization e.g. flow rate (m^3/h) in quantity (m^3).

Calculated values

The values of the universal inputs can be used to perform calculations in the mathematics channels.

Measuring range of analog universal inputs

According to IEC 60873-1: An additional display error of ±1 digit is permitted for every measured value.

User-definable measuring ranges per universal input of the multifunction card:

Measured variable	Measuring range	Maximum measured error of measuring range (oMR), temperature drift	Input resistance
Current (I)	0 to 20 mA; 0 to 20 mA quadratic 0 to 5 mA 4 to 20 mA; 4 to 20 mA quadratic ±20 mA Overrange: up to 22 mA or -22 mA	±0.1% oMR Temperature drift: ±0.01%/K oMR	Load: 50 Ω ±1 Ω
Voltage (U) >1 V	0 to 10 V; 0 to 10 V quadratic 0 to 5 V 1 to 5 V; 1 to 5 V quadratic ±10 V ±30 V	±0.1% oMR Temperature drift: ±0.01%/K oMR	≥1 MΩ
Voltage (U) ≤1 V	0 to 1 V; 0 to 1 V quadratic ±1 V ±150 mV	±0.1% oMR Temperature drift: ±0.01%/K oMR	≥2.5 MΩ
Resistance thermometer (RTD)	Pt100: -200 to 850 °C (-328 to 1562 °F) (IEC 60751:2008, α=0.00385) Pt100: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, α=0.003916) Pt100: -200 to 850 °C (-328 to 1562 °F) (GOST 6651-94, α=0.00391) Pt500: -200 to 850 °C (-328 to 1562 °F) (IEC 60751:2008, α=0.00385) Pt500: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, α=0.003916) Pt1000: -200 to 600 °C (-328 to 1112 °F) (IEC 60751:2008, α=0.00385) Pt1000: -200 to 510 °C (-328 to 950 °F) (JIS C 1604:1984, α=0.003916)	4-wire: ±0.1% oMR 3-wire: ±(0.1% oMR + 0.8 K) 2-wire: ±(0.1% oMR + 1.5 K) Temperature drift: ±0.01%/K oMR	
	Cu50: -50 to 200 °C (-58 to 392 °F) (GOST 6651-94, α=4260) Cu50: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, α=4280) Pt50: -200 to 1100 °C (-328 to 2012 °F) (GOST 6651-94, α=0.00391) Cu100: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, α=4280)	4-wire: ±0.2% oMR 3-wire: ±(0.2% oMR + 0.8 K) 2-wire: ±(0.2% oMR + 1.5 K) Temperature drift: ±0.02%/K oMR	

Measured variable	Measuring range	Maximum measured error of measuring range (oMR), temperature drift	Input resistance
	Pt46: -200 to 1100 °C (-328 to 2012 °F) (GOST 6651-94, α=0.00391) Cu53: -200 to 200 °C (-328 to 392 °F) (GOST 6651-94, α=4280)	4-wire: ±0.3% oMR 3-wire: ±(0.3% oMR + 0.8 K) 2-wire: ±(0.3% oMR + 1.5 K) Temperature drift: ±0.02%/K oMR	
Thermocouples (TC)	Type J (Fe-CuNi): -210 to 1200 °C (-346 to 2192 °F) (IEC 60584:2013) Type K (NiCr-Ni): -270 to 1300 °C (-454 to 2372 °F) (IEC 60584:2013) Type L (NiCr-CuNi): -200 to 800 °C (-328 to 1472 °F) (GOST R8.585:2001) Type L (Fe-CuNi): -200 to 900 °C (-328 to 1652 °F) (DIN 43710-1985) Type N (NiCrSi-NiSi): -270 to 1300 °C (-454 to 2372 °F) (IEC 60584:2013) Type T (Cu-CuNi): -270 to 400 °C (-454 to 752 °F) (IEC 60584:2013)	±0.1% oMR from -100 °C (-148 °F) ±0.1% oMR from -130 °C (-202 °F) ±0.1% oMR from -100 °C (-148 °F) ±0.1% oMR from -100 °C (-148 °F) ±0.1% oMR from -100 °C (-148 °F) ±0.1% oMR from -200 °C (-328 °F) Temperature drift: ±0.01%/K oMR	≥1 MΩ
	Type A (W5Re-W20Re): 0 to 2500 °C (32 to 4532 °F) (ASTME 988-96) Type B (Pt30Rh-Pt6Rh): 42 to 1820 °C (107.6 to 3308 °F) (IEC 60584:2013) Type C (W5Re-W26Re): 0 to 2315 °C (32 to 4199 °F) (ASTME 988-96) Type D (W3Re-W25Re): 0 to 2315 °C (32 to 4199 °F) (ASTME 988-96) Type R (Pt13Rh-Pt): -50 to 1768 °C (-58 to 3214 °F) (IEC 60584:2013) Type S (Pt10Rh-Pt): -50 to 1768 °C (-58 to 3214 °F) (IEC 60584:2013)	±0.15% oMR from 500 °C (932 °F) ±0.15% oMR from 600 °C (1112 °F) ±0.15% oMR from 500 °C (932 °F) ±0.15% oMR from 500 °C (932 °F) ±0.15% oMR from 100 °C (212 °F) ±0.15% oMR from 100 °C (212 °F) Temperature drift: ±0.01%/K oMR	≥1 MΩ
Pulse input (I) [1]	Min. Pulse length 40 µs, max. 12.5 kHz; 0 to 7 mA = LOW; 13 to 20 mA = HIGH		Load: 50 Ω ±1 Ω
Frequency input (I) [1]	0 to 10 kHz, overrange: up to 12.5 kHz; 0 to 7 mA = LOW; 13 to 20 mA = HIGH	±0.02% @ f <100 Hz of reading ±0.01% @ f ≥100 Hz of reading Temperature drift: 0.01% of measured value over the entire temperature range	

1) If a universal input is used as a frequency or pulse input, a series resistor must be used in series connection with the voltage source. Example: 1.2 kΩ series resistor at 24 V

Maximum load of inputs

Limit values for input voltage and current as well as cable open circuit detection/line influence/temperature compensation:

Measured variable	Limit values (steady-state, without destroying input)	Cable open circuit detection/line influence/temperature compensation
Current (I)	Maximum permitted input voltage: 2.5 V Maximum permitted input current: 50 mA	4 to 20 mA range with disengageable cable open circuit monitoring to NAMUR NE43. The following error ranges apply when NE43 is switched on: ≤3.8 mA: underrange ≥20.5 mA: overrange ≤ 3.6 mA or ≥ 21.0 mA: open circuit (display shows: – – – –)
Pulse, frequency (I)	Maximum permitted input voltage: 2.5 V Maximum permitted input current: 50 mA	No cable open circuit monitoring
Voltage (U) >1 V	Maximum permitted input voltage: 35 V	1 to 5 V range with disengageable cable open circuit monitoring: <0.8 V or >5.2 V: cable open circuit (display shows: - - - -)
Voltage (U) ≤1 V	Maximum permitted input voltage: 24 V	
Resistance thermometer (RTD)	Measuring current: ≤1 mA	Maximum barrier resistance (or line resistance): 4-wire: max. 200 Ω; 3-wire: max. 40 Ω Maximum influence of barrier resistance (or line resistance) for Pt100, Pt500 and Pt1000: 4-wire: 2 ppm/Ω, 3-wire: 20 ppm/Ω Maximum influence of barrier resistance (or line resistance) for Pt46, Pt50, Cu50, Cu53, Cu100 and Cu500: 4-wire: 6 ppm/Ω, 3-wire: 60 ppm/Ω Cable open circuit monitoring if any connection is interrupted.
Thermocouples (TC)	Maximum permitted input voltage: 24 V	Influence of line resistance: <0.001%/Ω Error, internal temperature compensation: ≤ 2 K

Scan rate

Current/voltage/pulse/frequency input: 100 ms per channel

Thermocouples and resistance temperature detector: 1 s per channel

Data storage/save cycle

Selectable save cycle. Choose from: 1s / 2s / 3s / 4s / 5s / 10s / 15s / 20s / 30s / 1min / 2min / 3min / 4min / 5min / 10min / 15min / 30min / 1h

Typical recording duration

Prerequisites for following tables:
- No limit value violation / integration
- Digital input not used
- Signal analysis 1: off, 2: day, 3: month, 4: year
- No active mathematics channels

Frequent entries in the event log reduce the memory availability!

128 MB internal memory:

Analog inputs	Channels in groups	Storage cycle (weeks, days, hours)				
		5 min	1 min	30 s	10 s	1 s
1	1/0/0/0	668, 4, 14	135, 0, 5	67, 4, 4	22, 3, 20	2, 1, 18
4	4/0/0/0	491, 0, 10	99, 4, 17	49, 6, 12	16, 4, 15	1, 4, 16
8	4/4/0/0	246, 1, 14	49, 6, 1	24, 6, 19	8, 2, 7	0, 5, 20
12	4/4/4/0	164, 2, 4	33, 1, 18	16, 4, 13	5, 3, 21	0, 3, 21

External memory, 1 GB SD card:

Analog inputs	Channels in groups	Storage cycle (weeks, days, hours)				
		5 min	1 min	30 s	10 s	1 s
~~1~~	~~1/0/0/0~~	~~12825, 5, 20~~	~~2580, 4, 18~~	~~1291, 2, 5~~	~~430, 4, 14~~	~~43, 0, 12~~
4	4/0/0/0	8672, 5, 12	1749, 6, 13	875, 6, 13	292, 1, 8	29, 1, 14
~~8~~	~~4/4/0/0~~	~~4343, 1, 1~~	~~875, 1, 17~~	~~438, 0, 6~~	~~146, 0, 17~~	~~14, 4, 7~~
~~12~~	~~4/4/4/0~~	~~2896, 6, 13~~	~~583, 3, 21~~	~~292, 0, 6~~	~~97, 2, 20~~	~~9, 5, 4~~

Converter resolution

24 bit

Totalization

The interim, daily, monthly and yearly value and the total value can be determined (13-digit, 64 bit).

Analysis

Recording of quantity/operating time (standard function), also a min/max/median analysis within the set time frame.

Digital inputs

Input level	To IEC 61131-2: logical "0" (corresponds to -3 to +5 V), activation with logical "1" (corresponds to +12 to +30 V)
Input frequency	max. 25 Hz
Pulse length	Min. 20 ms (pulse counter)
Pulse length	Min. 100 ms (control input, messages, operating time)
Input current	max. 2 mA
Input voltage	Max. 30 V

Selectable functions

- Functions of the digital input: control input, ON/OFF message, pulse counter (13-digit, 64 bit), operating time, message+operating time, quantity from time, Modbus slave.
- Functions of the control input: start recording, screen saver on, lock setup, time synchronization, limit monitoring on/off, lock keyboard/navigator, start/stop analysis.

Output

Auxiliary voltage output

The auxiliary voltage output can be used for loop power supply or to control the digital inputs. The auxiliary voltage is short-circuit proof and galvanically isolated.

Output voltage	24 V_{DC} ±15%
Output current	Max. 250 mA

Galvanic isolation

All inputs and outputs are galvanically isolated from each other and designed for the following testing voltages:

	Relay	Digital in	Analog in	Ethernet	RS232/RS485	USB	Auxiliary voltage output
Relay	500 V_{DC}	2 kV_{DC}	2 kV_{DC}	2 kV_{DC}	2 kV_{DC}	2 kV_{DC}	2 kV_{DC}
Digital in	2 kV_{DC}	Galvanically connected	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}
Analog in	2 kV_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}
Ethernet	2 kV_{DC}	500 V_{DC}	500 V_{DC}	-	500 V_{DC}	500 V_{DC}	500 V_{DC}
RS232/RS485	2 kV_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	-	500 V_{DC}	500 V_{DC}
USB	2 kV_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	Galvanically connected	500 V_{DC}
Auxiliary voltage output	2 kV_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	500 V_{DC}	-

Relay outputs

A mix of low voltage (230 V) and safety extra low voltage (SELV circuits) is not permitted at the connections of the relay contacts.

Alarm relay

1 alarm relay with changeover contact.

Standard relay

5 relays with NO contact, e.g. for limit value messages (can be configured as NC contact).

Relay switching capacity

- Max. switching capacity: 3 A @ 30 V DC
- Max. switching capacity: 3 A @ 250 V AC
- Min. switching load: 300 mW

Switching cycles

$>10^5$

Cable specification

Cable specification, spring terminals

All connections on the rear of the device are designed as pluggable screw or spring terminal blocks with reverse polarity protection. This makes the connection very quick and easy. The spring terminals are unlocked with a slotted screwdriver (size 0).

Please note the following when connecting:
- Wire cross-section, auxiliary voltage output, digital I/O and analog I/O: max. 1.5 mm^2 (14 AWG) (spring terminals)
- Wire cross-section, mains: max. 2.5 mm^2 (13 AWG) (screw terminals)
- Wire cross-section, relays: max. 2.5 mm^2 (13 AWG) (spring terminals)
- Stripping length: 10 mm (0.39 in)

No ferrules must be used when connecting flexible wires to spring terminals.

Shielding and grounding

Optimum electromagnetic compatibility (EMC) can only be guaranteed if the system components and, in particular, the lines - both sensor lines and communication lines - are shielded and the shield forms as complete a cover as possible. A shielded line must be used for sensor lines that are longer than 30 m. A shield coverage of 90% is ideal. In addition, make sure not to cross sensor lines and communication lines when routing them. Connect the shield as often as possible to the reference ground to ensure optimum EMC protection for the different communication protocols and the connected sensors.

To comply with requirements, three different types of shielding are possible:
- Shielding at both ends
- Shielding at one end on the supply side with capacitance termination at the device
- Shielding at one end on the supply side

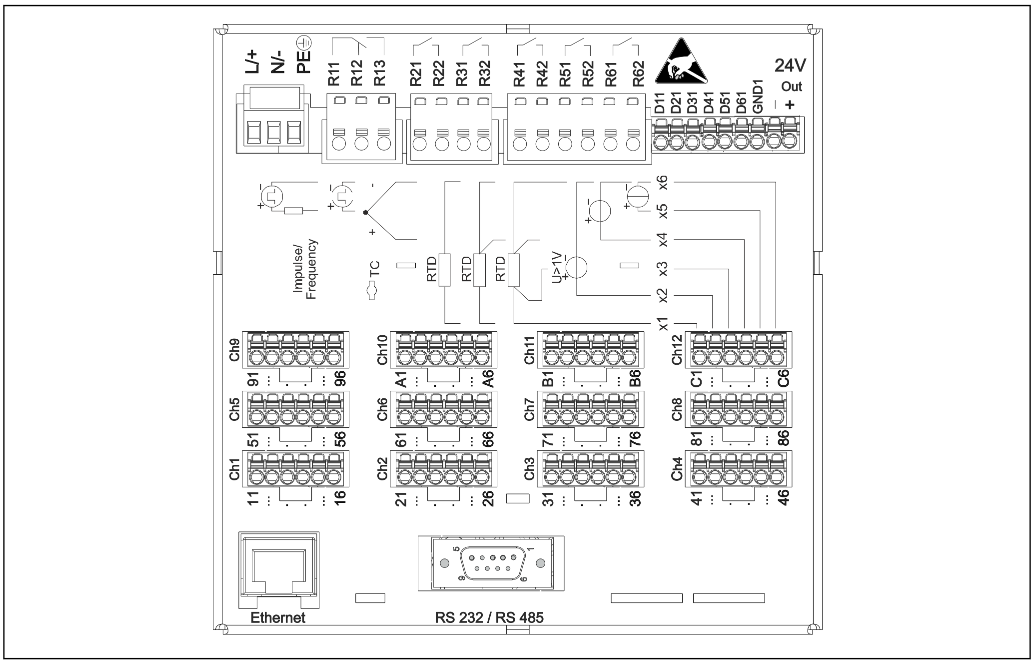
Experience shows that the best results with regard to EMC are achieved in most cases in installations with one-sided shielding on the supply side (without capacitance termination at the device). Appropriate internal device wiring measures must be taken to allow unrestricted operation when EMC interference is present. These measures have been taken into account for this device. Operation in the event of disturbance variables as per NAMUR NE21 is thus guaranteed.

Where applicable, national installation regulations and guidelines must be observed during the installation! Where there are large differences in potential between the individual grounding points, only one point of the shielding is connected directly with the reference ground.

If the shielding of the cable is grounded at more than one point in systems without potential matching, mains frequency equalizing currents can occur. These can damage the signal cable or significantly impact signal transmission. In such cases the shielding of the signal cable is to be grounded on one side only, i.e. it may not be connected to the ground terminal of the housing. The shield that is not connected should be insulated!

Power supply

Terminal assignment



1 *Terminals on back of device*

Supply voltage

- Extra-low voltage power supply unit ±24 V AC/DC (-10% / +15%) 50/60Hz
- ~~Low voltage power supply unit 100 to 230 V AC (±10%) 50/60Hz~~

An overload protection element (rated current ≤ 10 A) is required for the power cable.

Power consumption

- ~~100 to 230 V: max. 35 VA~~
- 24 V: max. 24 VA

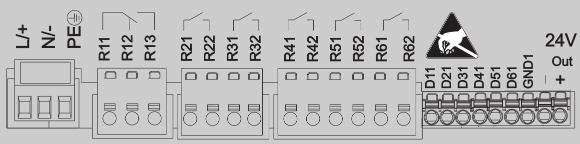
The power actually consumed depends on the individual operating state and the device version (LPS, USB, brightness of screen, number of channels, etc). The active power here is approx. 3 W to 20 W.

Power supply failure

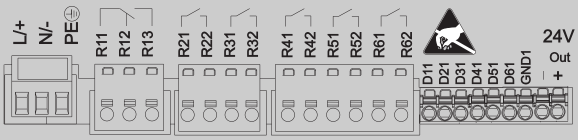
Battery-backed time and data memory. The device starts automatically following a power failure.

Electrical connection

Supply voltage

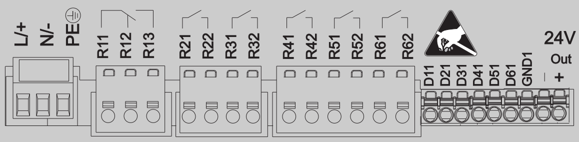
Power unit type	Terminal 		
~~100-230 VAC~~	~~L+~~	~~N-~~	~~PE~~
	~~Phase L~~	~~Zero conductor N~~	~~Ground~~
24 V AC/DC	L+	N-	PE
	Phase L or +	Zero conductor N or -	Ground

Relay

Type	Terminal (max. 250 V, 3 A)  A0019103				
Alarm relay 1	**R11**	**R12**	**R13**		
	Changeover contact	Normally closed contact (NC) [1]	Normally open contact (NO) [2]		
Relay 2 to 6				**Rx1**	**Rx2**
				Switching contact	Normally open contact (NO [2])

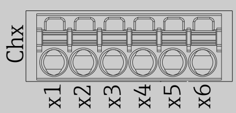
1) NC = normally closed (breaker)
2) NO = normally open (maker)

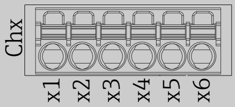
Digital inputs; auxiliary voltage output

Type	Terminal  A0019103			
Digital input 1 to 6	**D11 to D61**	**GND1**		
	Digital input 1 to 6 (+)	Ground (-) for digital inputs 1 to 6		
Auxiliary voltage output, not stabilized, max. 250 mA			**24V Out -**	**24V Out +**
			- Ground	+ 24V (±15%)

Analog inputs

The first digit (x) of the two-digit terminal number corresponds to the associated channel:

Type	Terminal  A0019303					
	x1	**x2**	**x3**	**x4**	**x5**	**x6**
Current/pulse/frequency input [1]					(+)	(-)
Voltage > 1V		(+)				(-)
Voltage ≤ 1V				(+)		(-)
Resistance thermometer RTD (2-wire)	(A)					(B)

Type	Terminal A0019303					
Resistance thermometer RTD (3-wire)	(A)			b (sense)		(B)
Resistance thermometer RTD (4-wire)	(A)		a (sense)	b (sense)		(B)
Thermocouples TC				(+)		(-)

1) If a universal input is used as a frequency or pulse input and the voltage is >2.5 V, a resistance must be used in series connection with the voltage source. Example: 1.2 kOhm series resistance at 24 V

Connector

- Panel-mounted device: connected to mains via plug-in screw terminals with reverse polarity protection
- Desktop version (option): connected to mains via IEC connector

Overvoltage protection

To avoid high-energy transients on long signal cables, connect a suitable surge arrester upstream (e.g. E+H HAW562) in series upstream.

Connection data interface, communication

USB ports (standard):

1 x USB port type A (host)

A USB 2.0 port is available on a shielded USB A socket at the front of the device. A USB stick, for example, can be connected to this interface as a storage medium. An external keyboard or USB hub may also be connected.

1 x USB port type B (function)

A USB 2.0 port is available on a shielded USB B socket at the front of the device. This can be used to connect the device for communication with a laptop, for example.

Ethernet interface (standard):

Ethernet interface on back, 10/100 Base-T, plug type RJ45. The Ethernet interface can be used to integrate the device via a hub or switch into a PC network (TCP/ IP Ethernet). A standard patch cable (e.g. CAT5E) can be used for the connection. Using DHCP, the device can be fully integrated into an existing network without the need for additional configuration. The device can be accessed from every PC in the network. Normally only the automatic assignment of the IP address must be configured at the client. When the device is started, it can automatically retrieve the IP address, subnet mask and gateway from a DHCP server. If a DHCP is not used, these settings must be made directly in the device (depends on the network to which the device is to be connected). Two Ethernet function LEDs are located on the rear of the device.

The following functions are implemented:
- Data communication with PC software (analysis software, configuration software, OPC server)
- Web server
- WebDAV (Web-based Distributed Authoring and Versioning) is an open standard for the provisioning of files via the HTTP protocol. The data saved on the device's SD card can be read out using a PC. A web browser or dedicated WebDAV client can be selected as network drive on the PC for this purpose.

Serial RS232/RS485 interface (option):

A combined RS232/RS485 connection is available on a shielded SUB D9 socket at the rear of the device. This can be used for data transfer and to connect a modem. For communication via modem, we recommend an industrial modem with a watchdog function.

- The following baud rates are supported: 9600, 19200, 38400, 57600, 115200
- Max. cable length with a shielded cable: 2 m (6.6 ft) (RS232), or 1000 m (3281 ft) (RS485)

Only one interface can be used at any one time (RS232 or RS485).

Performance characteristics

Response time

Input	Output	Time [ms]
Current, voltage, pulse	Relay	≤ 550
RTD	Relay	≤ 1150
TC [1]	Relay	≤ 1550
Cable open circuit detection, current input	Relay	≤ 1150
Sensor error RTD, TC	Relay	≤ 5000
Digital input	Relay	≤ 350

1) If internal measuring point temperature compensation is used, otherwise values as for voltage

Reference operating conditions

Reference temperature	25 °C (77 °F) ±5 K
Warm-up period	120 min.
Humidity	20 to 60 % rel. humidity

Hysteresis

Can be configured for limit values in the setup

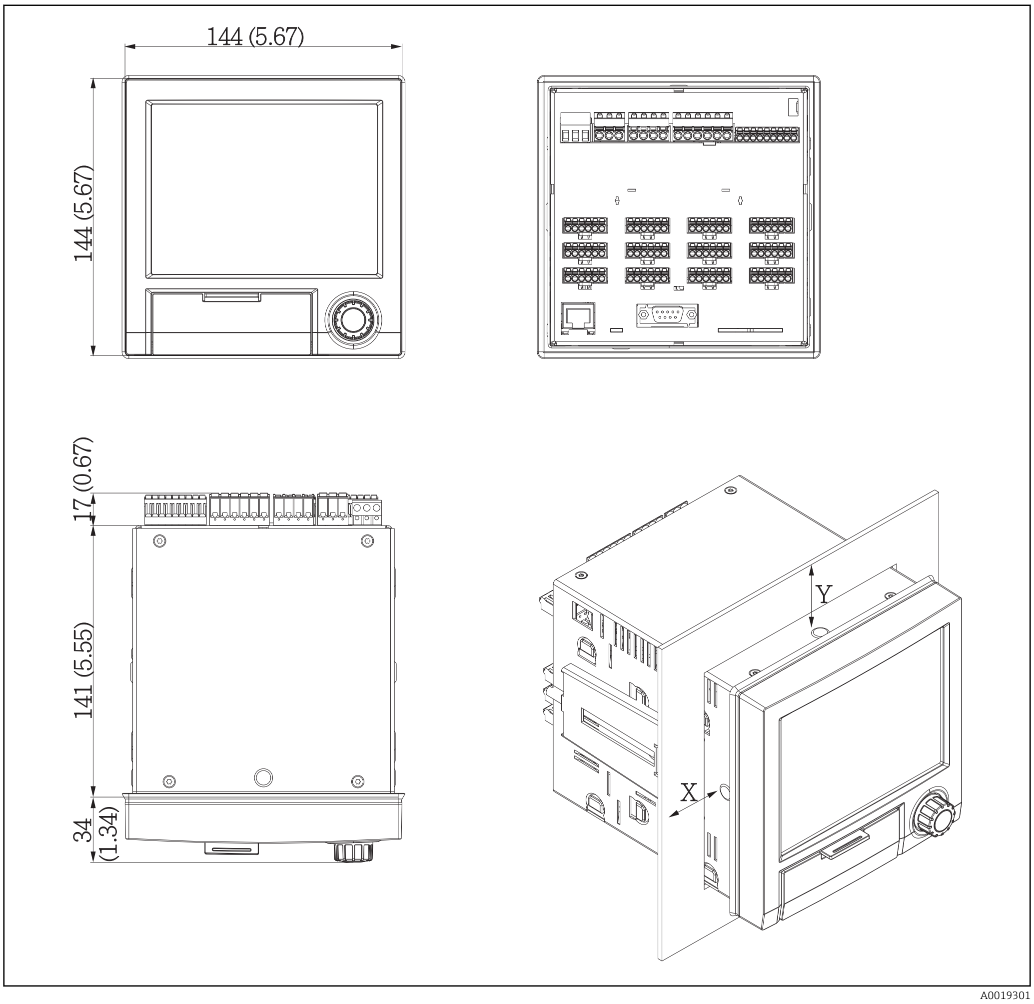
Long-term drift

As per IEC 61298-2: max. ±0.1%/year (of measuring range)

Installation

Mounting location and installation dimensions

The device is designed for use in a panel in non-hazardous areas.



2 *Panel mounting and dimensions in mm (in)*

Please observe the installation depth of approx. 158 mm (6.22 in) for the device incl. terminals and fastening clips.

- Panel cutout: 138 to 139 mm (5.43 to 5.47 in) x 138 to 139 mm (5.43 to 5.47 in)
- Panel strength: 2 to 40 mm (0.08 to 1.58 in)
- Angle of vision: from the midpoint axis of the display, 75° to the left and right, 65° above and below.
- A minimum distance of 15 mm (0.59 in) mm (inch) between the devices must be observed if aligning the devices in the Y-direction (vertically above one another). A minimum distance of 10 mm (0.39 in) mm (inch) between the devices must be observed if aligning the devices in the X-direction (horizontally beside one another).
- Securing to DIN 43 834

Field housing assembly and design (optional)

As an option, the device can be ordered ready-mounted in a field housing with IP65.

Dimensions (B x H x D) approx.: 320 mm (12.6 in) x 320 mm (12.6 in) x 254 mm (10 in)

Desktop housing assembly and design (optional)

As an option, the device can be ordered ready-mounted in a desktop housing.

Dimensions (B x H x D) approx.: 293 mm (11.5 in) x 188 mm (7.4 in) x 211 mm (8.3 in) (dimensions with bracket, feet and installed device)

Environment

Ambient temperature range

−10 to +50 °C (14 to 122 °F)

Storage temperature	−20 to +60 °C (−4 to +140 °F)
Humidity	5 to 85 %, non-condensing
Climate class	To IEC 60654-1: Class B2
Electrical safety	Class I equipment, overvoltage category II Pollution level 2
Altitude	< 2 000 m (6 561 ft) over MSL

Degree of protection

Front of panel-mounted device	IP65 / NEMA 4 (not assessed by UL)
Back of panel-mounted device (terminal side)	IP20

Electromagnetic compatibility

EMC in accordance with all relevant requirements of the IEC/EN 61326 series and NAMUR NE21. For details, refer to the Declaration of Conformity.

- Interference immunity: as per IEC/EN 61326 series (industrial environment) / NAMUR NE21
  Maximum measured error <1% of measuring range
- Interference emissions: as per IEC 61326-1 Class A

Mechanical construction

Design, dimensions

Information about design and dimensions → 📄 11

Weight

- Panel-mounted device with maximum configuration: approx. 2.2 kg (4.85 lbs)
- Desktop housing (excluding device): approx. 2.3 kg (5 lbs)
- Field housing (excluding device): approx. 4 kg (8.8 lbs)

Materials

Front frame	Zinc die cast GD-Z410, powder-coated
Sight glass	Transparent Makrolon plastic (FR clear 099) UL94-V2
Flap; jog/shuttle dial	Plastic ABS UL94-V2
Mounting guide rail for PCBs; motherboard fixing unit; display retainer plate	Plastic PA6-GF15 UL94-V2
Seal to panel wall; seal to display; seal in flap; seal to navigator	Rubber EPDM 70 Shore A
Casing; rear panel	Galvanized sheet steel St 12 ZE

All materials are silicone-free.

Materials of desktop housing

- Housing half-panels: sheet steel, electrolytically plated (powder-coated)
- Side sections: aluminum extruded section (powder-coated)
- Section ends: colored polyamide

Display and operating elements

Operating concept

The device can be operated directly onsite, or via remote configuration with the PC via interfaces and operating tools (Web server, configuration software).

Web server
A Web server is integrated into the device. The Web server offers the following range of functions:
- Easy configuration without additional installed software
- Instantaneous value display and diagnostics information
- Display of current measured value curves via Web browser (remote control)
- Display of historical measured data in numerical format or as a curve
- Display of events and logbook entries
- Loading/saving of device configuration
- Device firmware update
- Printout of device configuration

Integrated operating instructions
The device's simple operating concept enables you to perform commissioning for many applications without the need for hard copy operating instructions. The device has an integrated help function and displays operating instructions directly on screen if the navigator (jog/shuttle dial) is pressed for longer than 3 seconds.

Local operation

Display elements

Type

TFT color display

Size (diagonal screen measurement)

145 mm (5.7")

Resolution

VGA 307,200 pixels (640 x 480 pixels)

Backlight

70,000 h half value time (= half brightness)

Number of colors

262,000 viewable colors, 256 colors used

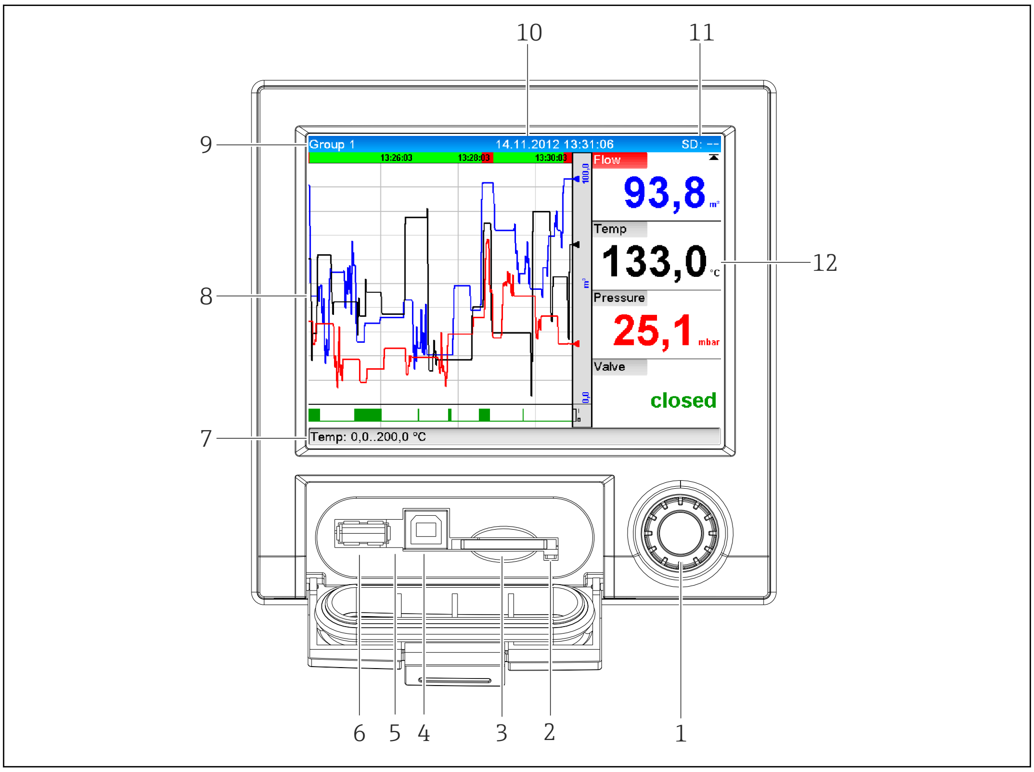
Viewing angle

Viewing angle: 130° vertical, 150° horizontal

Screen display

- Background color white
- Active channels can be assigned to up to 4 groups. These groups can be given a name e.g. "Temp. boiler 1" or "Daily averages" so that they can be uniquely identified.
- Linear scaling
- Measured value history: quick retrieval of historical data with zoom function
- Preformatted screen displays such as horizontal or vertical curve presentation, bar graph display or digital display.

Operating elements



3 *Front of device with open flap*

Item No.	Operating function (display mode = display of measured values) (Setup mode = operation in the Setup menu)
1	"Navigator": jog/shuttle dial for operating with additional press/hold function. In display mode: turn the dial to switch between the various signal groups. Press the dial to display the main menu. In setup mode or in a selection menu: turn the dial counterclockwise to move the bar or the cursor upwards or to the left, changes the parameter. Turning clockwise moves the bar or cursor down or clockwise, changes the parameter.
2	LED at SD slot. Orange LED lights up or flashes when the device writes to or reads from the SD card. **Do not remove the SD card if the LED is lit or flashing! Risk of data loss!**
3	Slot for SD card
4	USB B socket "Function" e.g. to connect to PC or laptop
5	Green LED lit: power supply present
6	USB A socket "Host" e.g. for USB memory stick or external keyboard
7	In display mode: alternating status display (e.g. set zoom range) of the analog or digital inputs in the appropriate color of the channel. In setup mode: different information can be displayed here depending on the display type.
8	In display mode: window for measured value display (e.g. curve display). In setup mode: display of operating menu
9	In display mode: current group name, type of evaluation In setup mode: name of the current operating item (dialog title)
10	In display mode: displays current date/time In setup mode: --

Item No.	Operating function (display mode = display of measured values) (Setup mode = operation in the Setup menu)
11	In display mode: alternating display indicating the percentage space on the SD card or USB stick that has already been used. Status symbols are also displayed in alternation with the memory information. In setup mode: the current "direct access" operating code is displayed
12	In display mode: display of current measured values and the status in the event of an error/alarm condition. In the case of counters, the type of counter is displayed as a symbol. If a measuring point has limit value status, the corresponding channel identifier is highlighted in red (quick detection of limit value violations). During a limit value violation and device operation, the acquisition of measured values continues uninterrupted.

Languages

The following languages can be selected in the operating menu: German, English, Spanish, French, Italian, Dutch, Swedish, Polish, Portuguese, Czech, Russian, Japanese, Chinese (Traditional), Chinese (Simplified)

Remote operation

Device access via operating tools

Device configuration and measured value retrieval can also be done via interfaces. The following operating tools are available for this purpose:

Operating tool	Functions	Access via
"Field Data Manager (FDM)" analysis software, SQL database support (included in the delivery)	▪ Export of saved data (measured values, analyses, event log) ▪ Visualization and processing of saved data (measured values, analyses, event log) ▪ Safe archiving of exported data in a SQL database	RS232/RS485, USB, Ethernet
Web server (integrated into the device; access via browser)	▪ Display of current and historical data and measured value curves via the Web browser ▪ Easy configuration without additional installed software ▪ Remote access to device and diagnostic information	Ethernet, or Ethernet via USB
OPC server (optional)	The following instantaneous values can be provided: ▪ Analog channels ▪ Digital channels ▪ Mathematics ▪ Totalizer	RS232/RS485, USB, Ethernet
"FieldCare / DeviceCare" configuration software	▪ Device configuration ▪ Loading and saving device data (upload/download) ▪ Documentation of the measuring point	USB, Ethernet

System integration

The device has (optional) fieldbus interfaces for exporting process values. Measured values and statuses can also be transmitted to the device via fieldbus. Alarms or errors in the context of data transmission are displayed depending on the bus system (e.g. status byte). The process values are transferred in the same units that are used for display at the device.

Ethernet

The following functions are implemented:
- Data communication with PC software (analysis software, configuration software, OPC server)
- Web server

~~Modbus RTU/TCP slave~~

~~The device can be connected to a Modbus system via RS485 or Ethernet interface. Up to 12 analog inputs and 6 digital inputs can be transmitted via Modbus and stored in the device.~~

Certificates and approvals

CE mark

The measuring system meets the legal requirements of the applicable EC guidelines. These are listed in the corresponding EC Declaration of Conformity together with the standards applied. The manufacturer confirms successful testing of the device by affixing to it the CE mark.

UL approval

UL recognized component (see www.ul.com/database, search for Keyword "E225237")

CSA

The product meets the requirements as per "CLASS 2252 05 - Process Control Equipment"

Other standards and guidelines

- IEC 60529:
  Degree of protection provided by housing (IP code)
- IEC/EN 61010-1:
  Safety requirements for electrical equipment for measurement, control and laboratory use
- IEC/EN 61326 Series:
  Electromagnetic compatibility (EMC requirements)

Ordering information

Ordering information

Detailed ordering information is available from the following sources:
- In the Product Configurator on the Endress+Hauser website: www.endress.com -> Click "Corporate" -> Select your country -> Click "Products" -> Select the product using the filters and search field -> Open product page -> The "Configure" button to the right of the product image opens the Product Configurator.
- From your Endress+Hauser Sales Center: www.addresses.endress.com

Product Configurator - the tool for individual product configuration
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Scope of delivery

The scope of delivery of the device comprises:
- Device (with terminals, as per order)
- Panel-mounted device: 2 screw fastening clips
- USB cable
- Optional: industrial grade SD card (card is located in the SD slot behind the flap on the front of the device)
- "Field Data Manager (FDM)" analysis software on DVD (Essential, Demo or Professional version, depending on order)
- Delivery note
- Multilanguage Brief Operating Instructions, hard copy

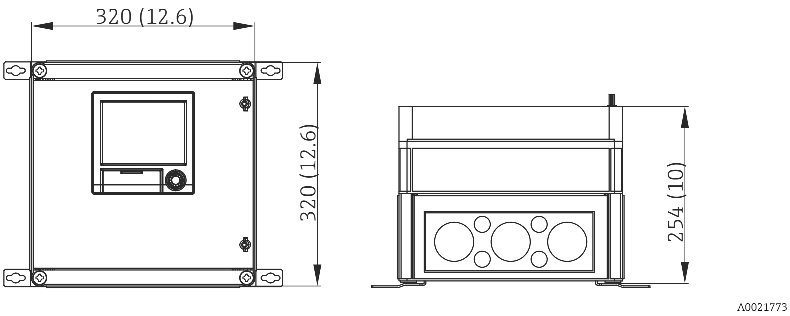
Accessories

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.

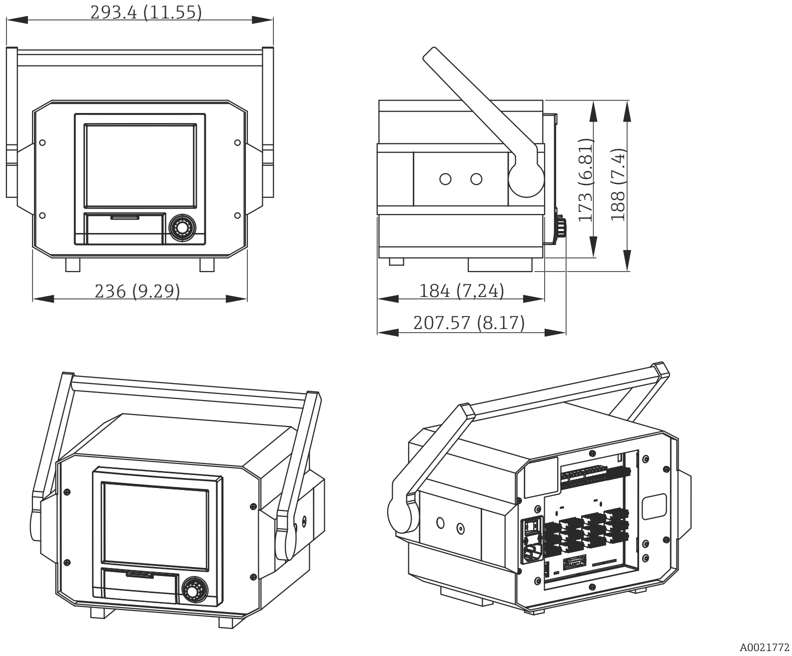
Device-specific accessories

Description	Order No.
"Industrial Grade" SD card, industry standard, 1GB	71213190
Field Data Manager analysis software with SQL database support (1 x workstation license, Professional version)	MS20-A1
OPC server software (full version on CD)	RXO20-11

Description	Order No.
Accessories for RXU10 data manager	RXU10- _ _
Designation: Cable set RS232 for connection to PC or modem USB - RS232 converter Cable USB-A - USB-B, 1.8 m (5.9 ft) Configuration software "FieldCare Device Setup" + USB cable	 RXU10-B _ RXU10-E _ RXU10-F _ RXU10-G _

Description	Order No.
Field housing IP65 (for panel-mounted device)  4 *Dimensions in mm (in)*	RXU10-H _

<table>
<tr><th>Description</th><th>Order No.</th></tr>
<tr><td>Desktop housing (for panel-mounted device), cable with Schuko plug
Desktop housing (for panel-mounted device), cable with US plug
Desktop housing (for panel-mounted device), cable with Swiss plug



5 *Dimensions in mm (in)*</td><td>RXU10-I _
RXU10-J _
RXU10-K _</td></tr>
<tr><td>Version:
Standard
Neutral</td><td>
RXU10- _ 1
RXU10- _ 2</td></tr>
</table>

Supplementary documentation

Standard documentation

- Technical Information for Ecograph T RSG35: TI01079R
- Operating Instructions for Ecograph T RSG35: BA01146R
- Brief Operating Instructions for Ecograph T RSG35: KA01132R
- System components and data manager - solutions to complete your measuring point: FA00016K

Supplementary device-dependent documentation

Operating Instructions for Ecograph T RSG35 with Modbus RTU / TCP slave: BA01258R

Data Sheet 4

Customer: Reference: Date:	Johnston Construction 32384 Charles County Pomonkey Pump Station Improvements 01/27/2026
<u>Qty</u>	<u>Description</u>
1	Mfg: Endress + Hauser: Digital Process Meter Display Model Number: RIA452-C211A11A Tags/Service: LEVEL DISPLAY / PLC/RTU Panel Process Meter Display Specifications: • RIA452 - Process indicator with pump control unit • C - FM AIS, I, II, III/2/ABCDEFG approvals • 2 - 20-30VDC, 20-28VAC power supply • 1 - 0/4-20mA measuring signal • 1 - 4 output relays, SPDT • A - Standard communication • 1 - 96x96 panel mounting housing, front IP65 • 1 - Basic version • A - Standard version

TI00113R/09/EN/14.19
71468732
2019-12-30

DIGITAL PROCESS METER DISPLAY

TAGS: LEVEL DISPLAY

Technical Information
RIA452

Process indicator

Digital process indicator in panel mounted housing for monitoring and displaying analog measured values with pump control and batch functions and flow calculation

Application

- Water/wastewater sector
- ~~Power industry~~
- ~~Raw materials~~
- ~~Chemicals industry~~
- ~~Food industry~~

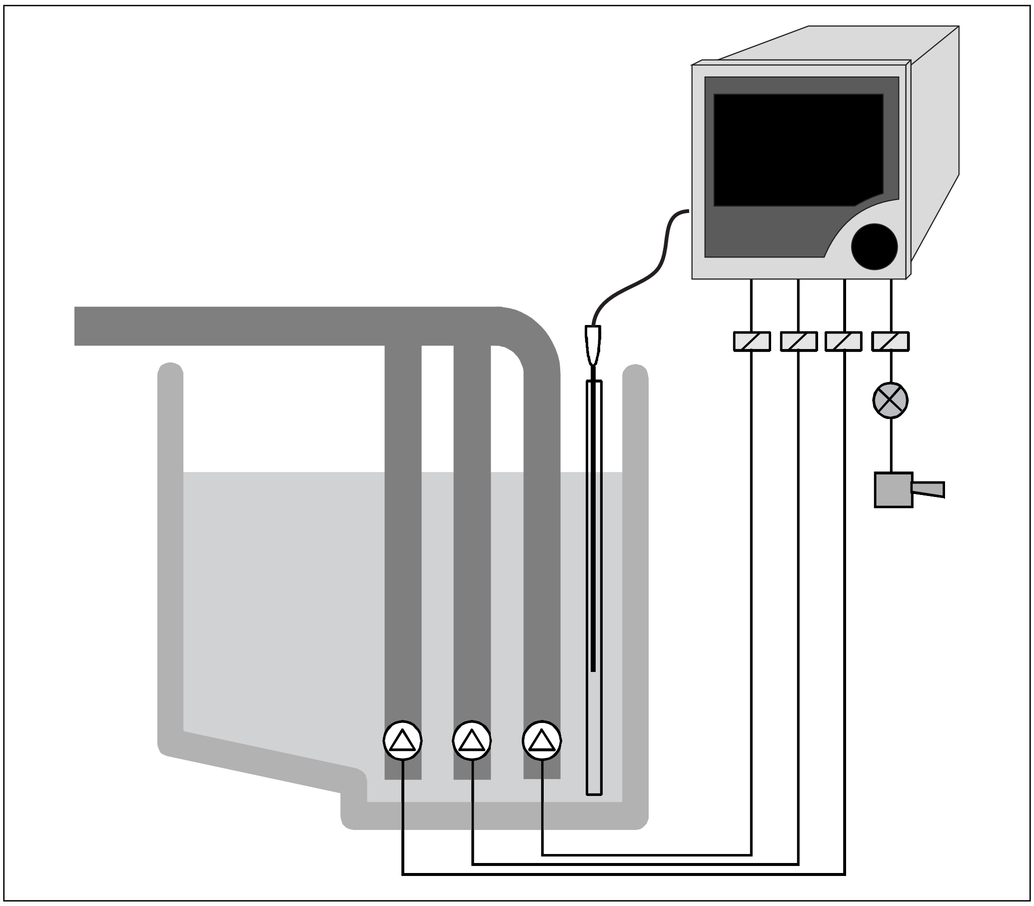
Your benefits

- 7-digit 14-segment LC display
- Multicolored
- Large bar graph with overrange and underrange
- Intrinsically safe input with transmitter power supply
- Digital status inputs for pump monitoring
- Universal input
- Up to eight relays
- Min/Max value saved
- Pump control functions
- Batch functions
- Flow measurement for open channels and weirs
- Linearization table with 32 support points
- Analog output
- Pulse output with totalizer
- Jog-shuttle operation
- Freely programmable units
- Configuration via interface and operating software
- Tank linearization via PC software

Endress+Hauser
People for Process Automation

Function and system design

Measuring principle



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1 *Example of an application of the process indicator*

The single-channel RIA452 process indicator monitors and displays analog measured values. Pumps can be monitored with the digital status inputs. The measured value is displayed using the seven-digit, 14-segment LC display. Numbers and units are displayed in white, the bar graph in yellow, overrange and underrange in red, and the limit value flags and digital status inputs in green and yellow. The RIA452 can provide power directly to two-wire transmitters that are connected. As an option, it is also possible to select the input and the transmitter power supply as intrinsically safe for Ex applications. Up to eight freely programmable relays monitor the measured value for limit value overshoot and undershoot. Other operating modes for the relays include sensor or device malfunction, batch and pump control functions (e.g. alternating pump control). Furthermore, the RIA452 can be used as a preset counter and for measuring flow at open channels and measuring weirs.

The scalable analog output offers many different ways of forwarding the input signal: zoom function, linearization, offset, inversion and signal conversion (input/output conversion). The optional pulse output allows users to output integrated process values.

Measuring system

Microcontroller-controlled indicator in a panel-mounted housing with a multicolored, illuminated LC display. Analog measured value acquisition is via an analog/digital converter. The digital status inputs are scanned cyclically. Power can be supplied directly to two-wire sensors with the transmitter power supply which is integrated as standard. The current input is optionally available as an intrinsically safe version for Ex applications. In this case, the RIA452 has a second, intrinsically safe transmitter power supply.

The freely scalable analog output is output via digital/analog conversion. The digital pulse output is output directly.

Up to eight relays are available in the device for monitoring limit values, pump control and batch functions.

The device can be operated either onsite via the jog/shuttle dial, or via the PC with an operating software. Operation can be locked using the hardware key or software code.

Linearization

The following flow curves are programmed into the device for open channels and weirs:
- Khafagi-Venturi flume
- ISO Venturi flume
- BST [1] Venturi flume
- Parshall flume
- Palmer-Bowlus flume
- Rectangular weir
- Rectangular weir with constriction
- NFX [2] rectangular weir
- NFX [2] rectangular weir with constriction
- Trapezoidal weir
- Triangular weir
- BST [1] triangular weir
- NFX [2] triangular weir

User-configurable flow formula

$Q = C * (h^{\alpha} + \gamma * h^{\beta})$

The parameters α, β, γ and C can be entered freely.

Linearization function

Up to 32 user-definable linearization points are available in the device for the linearization of the input, e.g. for tank linearization.

The linearization table for standard tanks and customer-specific tanks can be generated with the ReadWin 2000 operating software.

Input

Measured variable

- Current (standard)
- Digital inputs (standard)
- Current/voltage, resistance, RTD assembly, thermocouples (universal input option)

Measuring range

Current input:

Current:
- 0/4 to 20 mA +10% overrange, 0 to 5 mA
- Short-circuit current: max. 150 mA
- Input impedance: ≤ 5 Ω
- Response time: ≤ 100 ms

Universal input:

Current:
- 0/4 to 20 mA + 10% overrange, 0 to 5 mA
- Short-circuit current: max. 100 mA
- Input impedance: ≤ 50 Ω

Voltage:
- ±150 mV, ±1 V, ±10 V, ±30 V, 0 to 100 mV, 0 to 200 mV, 0 to 1 V, 0 to 10 V
- Input impedance: ≥ 100 kΩ

Resistance:
30 to 3 000 Ω in 3/4-wire technology

RTD assembly:
- Pt100/500/1000, Cu50/100, Pt50 in 3/4-wire technology
- Measuring current for Pt100/500/1000 = 0.25 mA

1) BST: British Standard
2) NFX: French standard NFX 10-311

Thermocouple types:
- J, K, T, N, B, S, R as per IEC584
- D, C as per ASTME998
- U, L as per DIN43710/GOST
- Response time: ≤ 100 ms

Digital input:

Digital input:
- Voltage level −3 to 5 V low, 12 to 30 V high (as per DIN19240)
- Input voltage max. 34.5 V
- Input current typ. 3 mA with overload and reverse polarity protection
- Sampling frequency max. 10 Hz

Galvanic isolation

Towards all other circuits

Output

Output signal

- Relay, transmitter power supply (standard)
- Current, voltage, pulse, intrinsically safe transmitter power supply (option)

Signal on alarm

No measured value visible on the LC display, no background illumination, no sensor power supply, no output signals, relays behave in safety-oriented manner.

Current/voltage output

Analog output range:
0/4 to 20 mA (active), 0 to 10 V (active)

Load:
- ≤ 600 Ω (current output)
- Max. output current 22 mA (voltage output)

Signal characteristics:
Freely scalable signal

Galvanic isolation towards all other circuits

Pulse output (open collector)

Pulse output (open collector):
- Frequency range to 2 kHz
- I_{max} = 200 mA
- U_{max} = 28 V
- $U_{low/max}$ = 2 V at 200 mA
- Pulse width = 0.04 to 2 000 ms

Relay output

Signal characteristics:
Binary, switches when the limit value is reached

Switch function: limit relay switches for the operating modes:
- Minimum/maximum safety
- Alternating pump control function
- Batch function
- Time control
- Window function
- Gradient
- Device malfunction
- Sensor malfunction

Switching threshold:
Freely programmable

Hysteresis:
0 to 99%

Signal source:
- Analog input signal
- Integrated value
- Digital input

Number:
4 in basic unit ~~(can be extended to 8 relays, option)~~

Electrical specifications:
- Relay type: changeover
- Relay switching capacity: 250 V_{AC} / 30 V_{DC}, 3 A
- Switch cycles: typically 10^5
- Switching frequency: max. 5 Hz
- Minimum switching load: 10 mA / 5 V_{DC}

Galvanic isolation towards all other circuits

Mixed assignment of low and extra-low voltage circuits is not permitted for neighboring relays.

Transmitter power supply

Transmitter power supply 1, terminal 81/82 (optionally intrinsically safe):

Electrical specifications:
- Output voltage: 24 V ±15%
- Output current: max. 22 mA (for U_{out} ≥ 16 V, sustained short-circuit proof)
- Impedance: ≤ 345 Ω

Transmitter power supply 2, terminal 91/92:

Electrical specifications:
- Output voltage: 24 V ±15%
- Output current: max. 250 mA (sustained short-circuit proof)

Transmitter power supply 1 and 2:

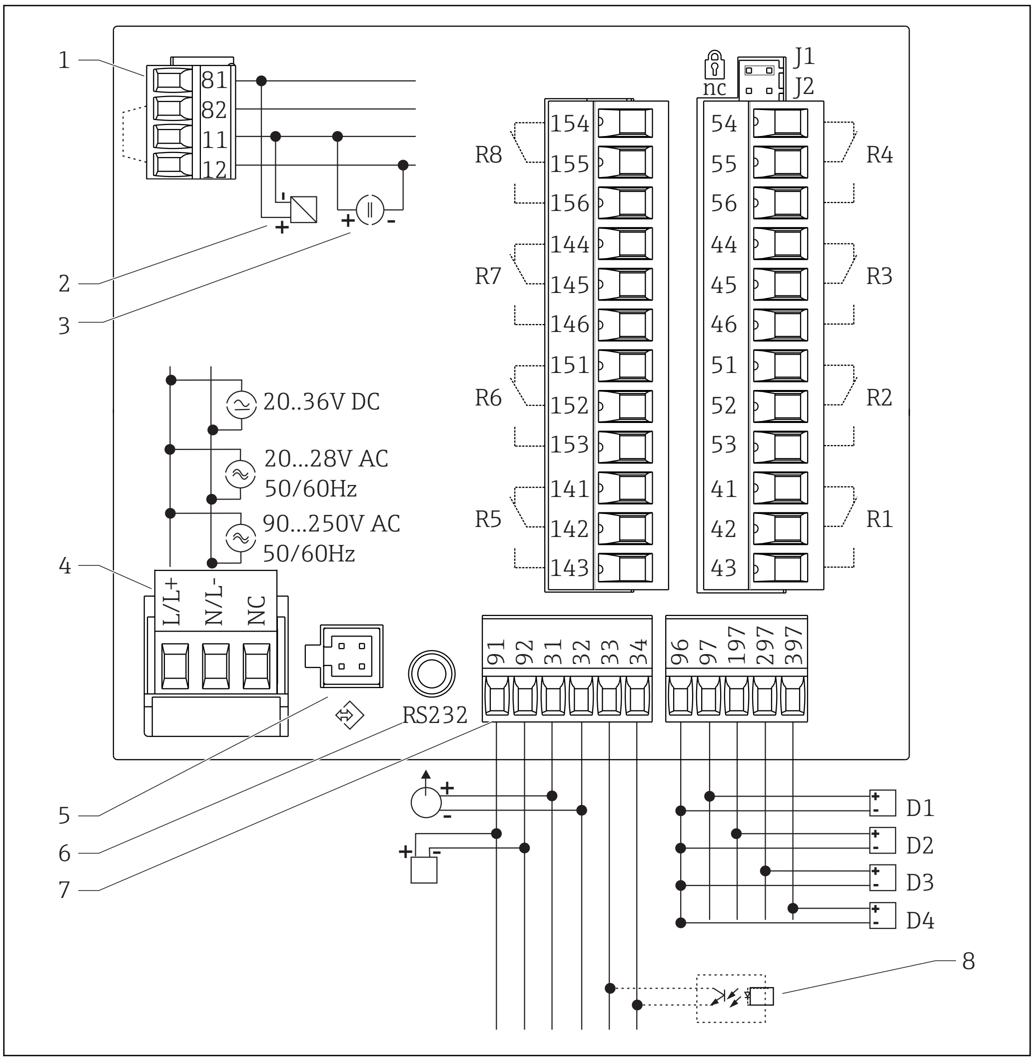
Galvanic isolation:
Towards all other circuits

HART®

HART® signals are not affected

Power supply

Terminal assignment



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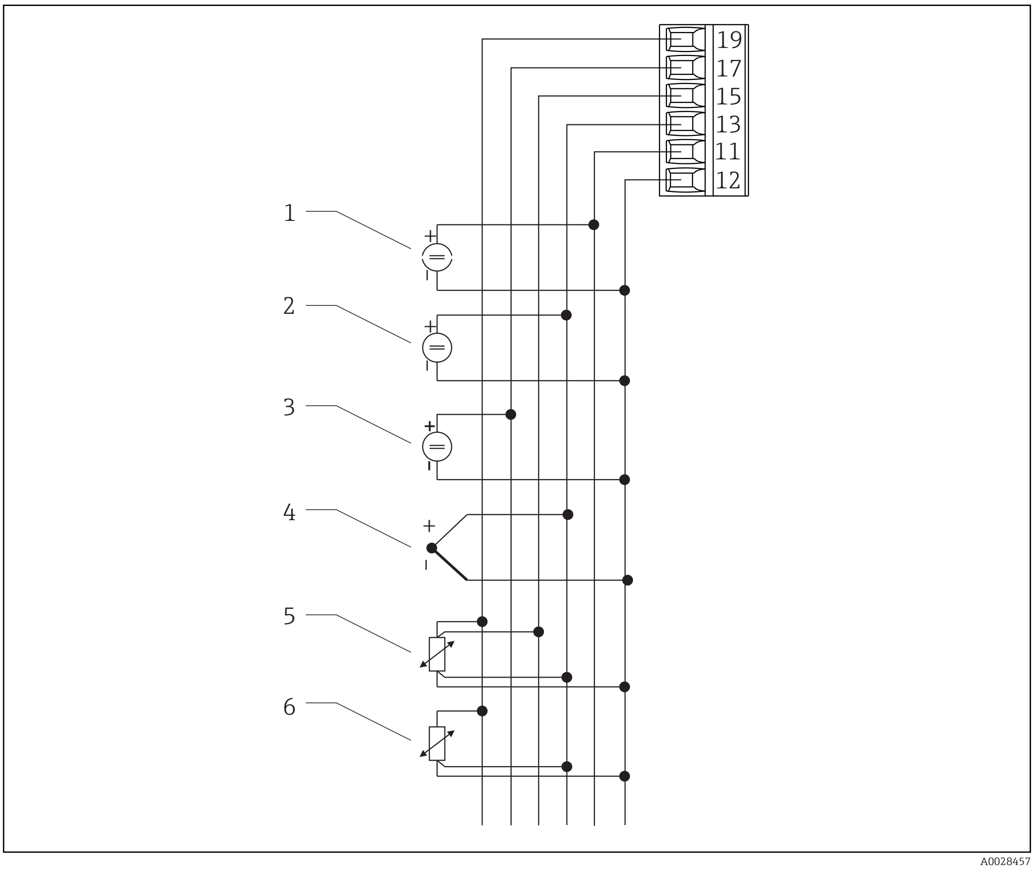
2 *Terminal assignment of process indicator*

1	*Current input (12 and 82 jumpered internally)*	*7*	*Transmitter power supply and analog output*
2	*- passive sensor*	*8*	*Open collector output*
3	*- active sensor*	*D1 to D4*	*Digital inputs*
4	*Power supply*	*R1 to R4*	*Relay outputs*
5	*Interface for PC operating software*	*R5 to R8*	*Relay outputs (optional)*
6	*RS232 interface*	*J1*	Hardware write protection

Universal input option



3 *Universal input terminal assignment*

1 *Current input 0/4 to 20 mA*
2 *Voltage input ±1 V*
3 *Voltage input ±30 V*
4 *Thermocouples*
5 *RTD assembly, 4-wire*
6 *RTD assembly, 3-wire*

Interface connection data

RS232
- Connection: jack socket 3.5 mm, rear of device
- Transmission protocol: ReadWin 2000
- Transmission rate: 38 400 Baud

Supply voltage

- Low voltage power unit 90 to 250 V_{AC}50/60 Hz
- Extra-low voltage power unit 20 to 36 V_{DC} or 20 to 28 V_{AC} 50/60 Hz
  The device must be powered only by a power unit that operates using a limited energy circuit in accordance with UL/EN/IEC 61010-1, Section 9.4 and the requirements in Table 18.

Power consumption

Power consumption max. 24 VA

Performance characteristics

Reference operating conditions

Power supply: 230 V_{AC} ±10%, 50 Hz ±0.5 Hz

Warm-up period: 90 min

Ambient temperature: 25 °C (77 °F)

Maximum measured error

Current input

Accuracy	0.1% of full scale
Resolution	13 bit
Temperature drift	≤ 0.4%/10 K (18 °F)

Universal input

	Input:	Range:	Maximum measured error of measuring range (oMR):
Accuracy	Current	0 to 20 mA, 0 to 5 mA, 4 to 20 mA; overrange: to 22 mA	±0.10%
	Voltage > 1 V	0 to 10 V, ±10 V, ±30 V	±0.10%
	Voltage ≤ 1 V	±1 V, 0 to 1 V, 0 to 200 mV, 0 to 100 mV, ±150 mV	±0.10%
	Resistance thermometer	Pt100, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604, GOST) Pt500, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604) Pt1000, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604)	4-wire: ± (0.10% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.15% oMR + 0.8 K (1.44 °F))
		Cu100, −200 to 200 °C (−328 to 392 °F) (GOST) Cu50, −200 to 200 °C (−328 to 392 °F) (GOST) Pt50, −200 to 600 °C (−328 to 1 112 °F) (GOST)	4-wire: ± (0.20% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.20% oMR + 0.8 K (1.44 °F))
	Resistance measurement	30 to 3 000 Ω	4-wire: ± (0.20% oMR + 0.3 K (0.54 °F) 3-wire: ± (0.20% oMR + 0.8 K (1.44 °F))
	Thermocouples	Typ J (Fe-CuNi), −210 to 999.9 °C (−346 to 1 382 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ K (NiCr-Ni), −200 to 1 372 °C (−328 to 2 502 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −130 °C (−234 °F)
		Typ T (Cu-CuNi), −270 to 400 °C (−454 to 752 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −200 °C (−328 °F)
		Typ N (NiCrSi-NiSi), −270 to 1 300 °C (−454 to 2 372 °F) (IEC584)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ B (Pt30Rh-Pt6Rh), 0 to 1 820 °C (32 to 3 308 °F) (IEC584)	± (0.15% oMR + 1.5 K (2.7 °F)) from 600 °C (1 112 °F)
		Typ D (W3Re/W25Re), 0 to 2 315 °C (32 to 4 199 °F) (ASTME998)	± (0.15% oMR + 1.5 K (2.7 °F)) from 500 °C (932 °F)
		Typ C (W5Re/W26Re), 0 to 2 315 °C (32 to 4 199 °F) (ASTME998)	± (0.15% oMR + 1.5 K (2.7 °F)) from 500 °C (932 °F)
		Typ L (Fe-CuNi), −200 to 900 °C (−328 to 1 652 °F) (DIN43710, GOST)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ U (Cu-CuNi), −200 to 600 °C (−328 to 1 112 °F) (DIN43710)	± (0.15% oMR + 0.5 K (0.9 °F)) from −100 °C (−148 °F)
		Typ S (Pt10Rh-Pt), 0 to 1 768 °C (32 to 3 214 °F) (IEC584)	± (0.15% oMR + 3.5 K (6.3 °F)) for 0 to 100 °C (32 to 212 °F) ± (0.15% oMR + 1.5 K (2.7 °F)) for 100 to 1 768 °C (212 to 3 214 °F)
		Typ R (Pt13Rh-Pt), −50 to 1 768 °C (−58 to 3 214 °F) (IEC584)	± (0.15% oMR + 1.5 K (2.7 °F)) for 100 to 1 768 °C (212 to 3 214 °F)
Resolution		16 bit	
Temperature drift		Temperature drift: ≤ 0.1%/10 K (18 °F)	

Current output

Linearity	0.1% of full scale
Resolution	13 bit
Temperature drift	Temperature drift: ≤ 0.1%/10 K (18 °F)
Output Ripple	10 mV at 500 Ω for frequencies ≤ 50 kHz

Voltage output

Linearity	0.1% of full scale
Resolution	13 bit
Temperature drift	Temperature drift: ≤ 0.1%/10 K (18 °F)

Installation

Mounting location

Panel, cut-out 92 x 92 mm (3.62x3.62 in) (see 'Mechanical construction').

Orientation

Horizontal +/- 45° in every direction

Environment

Ambient temperature range

−20 to 60 °C (−4 to 140 °F)

Storage temperature

−30 to 70 °C (−22 to 158 °F)

Operating altitude

< 3 000 m (9 840 ft) over MSL

Climate class

To IEC 60654-1, Class B2

Degree of protection

Front IP 65 / NEMA 4

Device casing IP 20

Shock and vibration resistance

2 Hz (+3/-0) ... 13.2 Hz: ±1 mm (±0.04 in)

13.2 to 100 Hz: 0.7 g

Electromagnetic compatibility (EMC)

CE compliance

Electromagnetic compatibility in accordance with all the relevant requirements of the IEC/EN 61326 series and NAMUR Recommendation EMC (NE21). For details refer to the EU Declaration of Conformity.

Maximum measured error <1% of measuring range.

Interference immunity as per IEC/EN 61326 series, industrial requirements.

Interference emission as per IEC/EN 61326 series, Class A equipment.

Electrical protection class

IEC 60529 (IP code) / NEMA 250

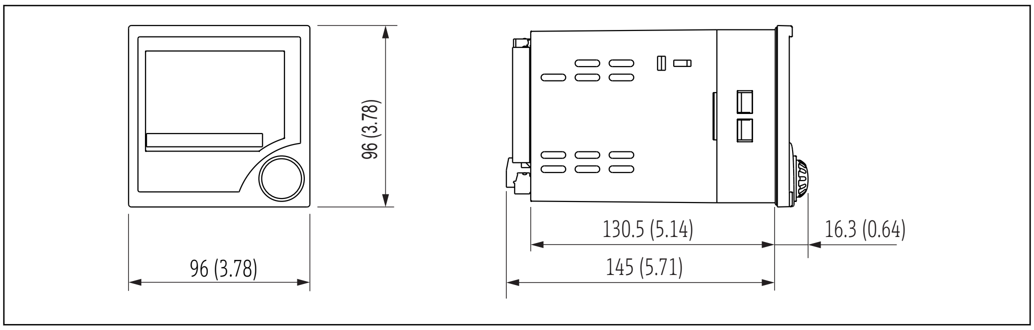
Condensation

Front: permitted

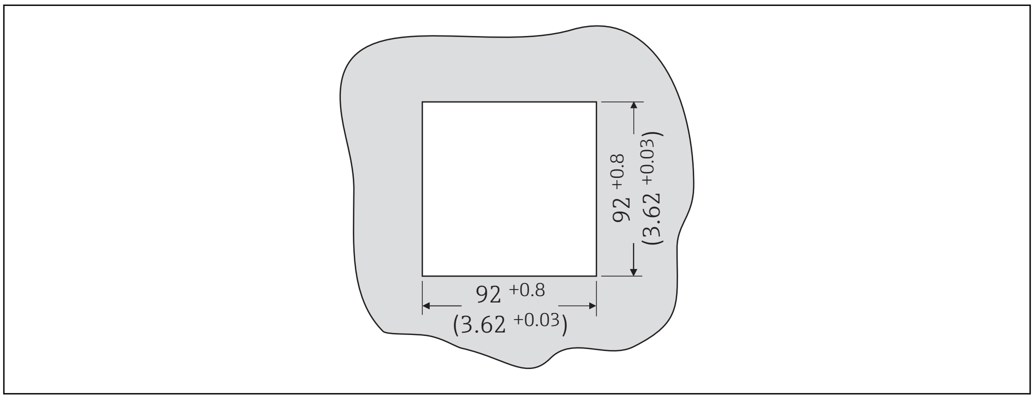
Device casing: not permitted

Mechanical construction

Design, dimensions



4 *Dimensions of the process indicator in mm (in)*



5 *Panel cutout, dimensions in mm (in)*

Weight

500 g (17.64 oz)

Materials

- Housing front: ABS plastic
- Housing casing: ABS GF plastic

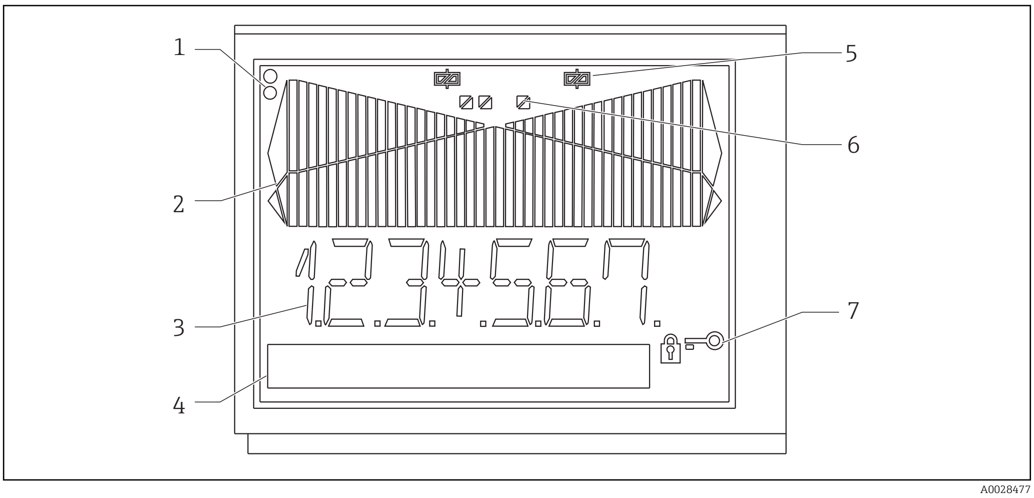
Terminals

Plug-in screw terminals, clamping range 1.5 mm² (16 AWG) solid, 1 mm² (18 AWG) strand with wire ferrule

Operability

Local operation

Display elements



6 Display elements of the process indicator

1 *Device status LEDs: green - device ready for operation; red - device or sensor malfunction*
2 *Bar graph with overrange and underrange*
3 *7-digit 14-segment display*
4 *Unit and text field 9x77 dot matrix*
5 *Relay status indicator: if power is supplied to a relay, the symbol is displayed*
6 *Status indicator for digital inputs*
7 *Symbol for "device operation locked"*

- Display range
  - -99999 to +99999 for measured values
  - 0 to 9999999 for counter values
- Signalization
  - Relay activation
  - Overrange/underrange

Operating elements

Jog/shuttle dial

Remote operation

Configuration

The device can be configured with the ReadWin 2000 PC software.

Interface

CDI interface at device; connection to PC via USB box (see "Accessories")

RS232 interface at device; connection with serial interface cable (see "Accessories")

Certificates and approvals

CE mark	The product meets the requirements of the harmonized European standards. As such, it complies with the legal specifications of the EC directives. The manufacturer confirms successful testing of the product by affixing to it the CE-mark.
Ex approval	Information about currently available Ex versions (ATEX, FM, CSA. etc.) can be supplied by your Endress+Hauser sales organization on request. All explosion protection data are given in a separate documentation which is available upon request.
Other standards and guidelines	The manufacturer confirms compliance with all the relevant external standards and guidelines.

Ordering information

Detailed ordering information is available for your nearest sales organization www.addresses.endress.com or in the Product Configurator under www.endress.com :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.

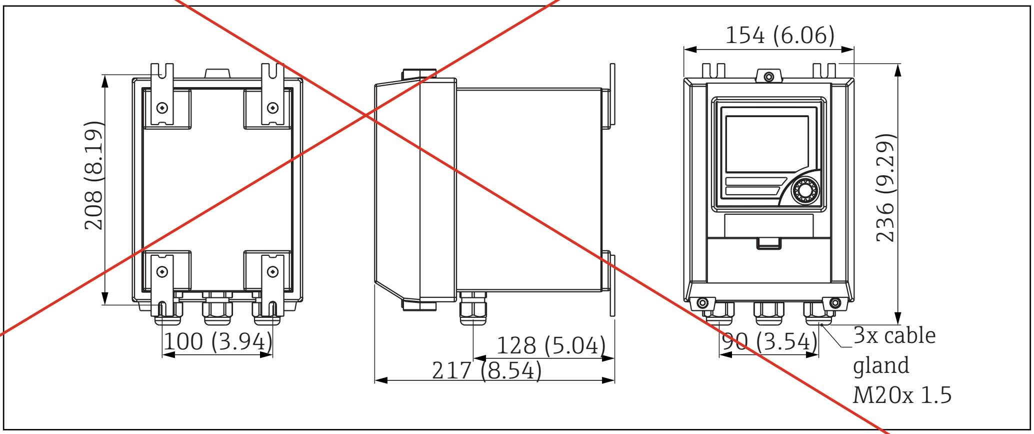
Product Configurator - the tool for individual product configuration
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

Accessories

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: www.endress.com.

Device-specific accessories

Designation	Order No.
PC configuration software ReadWin 2000 and serial configuration cable with 3.5 mm jack plug for RS232 port	RIA452A-VK
PC configuration software ReadWin 2000 and serial configuration cable for USB port with CDI connector	TXU10-AA
Field housing in IP65 → 7, 13	51009957
Current simulator active 4-20mA 1-channel, compact housing, 9V block battery	SONDST-S1



7 *Field housing dimensions*

Supplementary documentation

- System components and data manager - solutions to complete your measuring point: FA00016K/09
- Brief Operating Instructions for process indicator RIA452: KA00264R/09
  Operating Instructions for process indicator RIA452: BA00265R/09
- Ex-related additional documentation:
  ATEX II(1)GD: XA00053R/09/a3